

Properties of Triangles

Question1

In $\triangle ABC$, if $C = 120^\circ$, $c = \sqrt{19}$ and $b = 3$, then $a =$

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Options:

A.

4

B.

5

C.

2

D.

$\sqrt{5}$

Answer: C

Solution:

$$b = 3, c = \sqrt{19} \text{ and } \angle C = 120^\circ$$

$$\therefore c^2 = a^2 + b^2 - 2ab \cos C$$

$$\Rightarrow 19 = a^2 + 9 - 6a \cdot \cos 120^\circ$$

$$\Rightarrow 19 = a^2 + 9 + 3a$$

$$\Rightarrow a^2 + 3a - 10 = 0$$

$$\Rightarrow a^2 + 5a - 2a - 10 = 0$$

$$\Rightarrow (a + 5)(a - 2) = 0$$

$$\therefore a \neq -5$$

Hence, $a = 2$

Question2

In a $\triangle ABC$, $2A + C = 300^\circ$. If the circumradius of the $\triangle ABC$ is eight times its inradius, then $\sin \frac{C}{2} =$

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Options:

A.

$$\frac{1}{2}$$

B.

$$\frac{1}{4}$$

C.

$$\frac{3}{4+\sqrt{3}}$$

D.

$$\frac{1}{\sqrt{2}+1}$$

Answer: B

Solution:

$$\therefore 2A + C = 300^\circ \quad \dots (i)$$

$$\text{and } A + B + C = 180^\circ \quad \dots (ii)$$

$$\begin{aligned} \Rightarrow (2A + C) - (A + B + C) &= 300^\circ - 180^\circ = 120^\circ \\ \Rightarrow A - B &= 120^\circ \Rightarrow A = B + 120^\circ \end{aligned}$$

By Eq. (i)

$$2B + C = 60^\circ \Rightarrow C = 60^\circ - 2B$$

given that $R = 8r$

$$\text{and } \frac{r}{R} = 4 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right)$$

$$\Rightarrow \frac{1}{8} = 4 \sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{B}{2}\right) \cdot \sin\left(\frac{C}{2}\right)$$

substitute values

$$\frac{1}{8 \times 4} = \sin\left(\frac{B}{2} + 60^\circ\right) \sin\left(\frac{B}{2}\right) \sin(30^\circ + B)$$

let $B/2 = u$

$$\frac{1}{32} = \left(\frac{1}{2} \sin u + \frac{\sqrt{3}}{2} \cos u\right)$$

$$\sin u \left(\frac{1}{2} \cos 2u - \frac{\sqrt{3}}{2} \sin 2u\right)$$

$$\Rightarrow \frac{1}{32} = \frac{1}{4} \sin u (\sin u + \sqrt{3} \cos u)$$

$$(\cos 2u - \sqrt{3} \sin 2u)$$

$$\Rightarrow \frac{1}{32} = \frac{1}{4} \sin u \cdot (2 \sin(u + 60^\circ)) \cdot 2 \cos(2u + 60^\circ)$$

$$\Rightarrow \frac{1}{32} = \sin u \cdot \sin(u + 60^\circ) \cdot \cos(2u + 60^\circ)$$

$$\Rightarrow \frac{1}{32} = \left[\frac{1}{2} (\cos(u - u - 60^\circ) - \cos(u + u + 60^\circ))\right] \cdot \cos(2u + 60^\circ)$$

$$\Rightarrow \frac{1}{32} = \left[\frac{1}{4} - \frac{1}{2} \cos(2u + 60^\circ)\right] \cdot \cos(2u + 60^\circ)$$

let $\cos(2u + 60^\circ) = t$

$$\therefore \frac{1}{32} = \left(\frac{1}{4} - \frac{1}{2}t\right)t$$

$$\Rightarrow \frac{1}{32} = \frac{t}{4} - \frac{t^2}{2}$$

$$\Rightarrow \frac{1}{8} = t - 2t^2$$

$$\Rightarrow 16t^2 - 8t + 1 = 0$$

$$\therefore t = \frac{1}{4}$$

$$\Rightarrow \cos(2u + 60^\circ) = \frac{1}{4}$$

put $u = B/2$

$$\Rightarrow \cos(B + 60^\circ) = \frac{1}{4}$$

Now, $C/2 = 30^\circ - B$

and

$$\sin C/2 = \sin(30^\circ - B) = \cos(90^\circ - (130^\circ - B))$$

Hence, $\sin C/2 = 1/4$

Question3

In $\triangle ABC$, if $a = 5$, $b = 4$ and $\cos(A - B) = \frac{31}{32}$, then $c =$

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Options:

A.

8

B.

$\sqrt{41}$

C.

6

D.

$\sqrt{24}$

Answer: C

Solution:

$$\because \cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\begin{aligned} \text{and } \cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{16 + c^2 - 25}{2 \times 4 \times c} = \frac{c^2 - 9}{8c} \\ \cos B &= \frac{a^2 + c^2 - b^2}{2ac} = \frac{c^2 + 9}{10c} \end{aligned}$$

$$\text{also, } \sin A = \sqrt{1 - \cos^2 A} \text{ and}$$

$$\sin B = \sqrt{1 - \cos^2 B}$$

$$\begin{aligned} \Rightarrow \frac{31}{32} &= \left(\frac{c^2 - 9}{8c} \right) \left(\frac{c^2 + 9}{10c} \right) \\ &+ \left(\sqrt{1 - \left(\frac{c^2 - 9}{8c} \right)^2} \right) \left(\sqrt{1 - \left(\frac{c^2 + 9}{10c} \right)^2} \right) \end{aligned}$$

from the options

put $C = 6$

$$\cos A = \frac{36 - 9}{8 \times 6} = 9/16 \text{ and}$$

$$\cos B = \frac{36 + 9}{10 \times 6} = 3/4$$

$$\sin A = \sqrt{1 - \left(\frac{9}{16}\right)^2} = \frac{\sqrt{175}}{16}$$

$$\sin B = \sqrt{1 - (3/4)^2} = \sqrt{7}/4$$

$$\begin{aligned} \therefore \frac{9}{16} \times \frac{3}{4} + \frac{\sqrt{175}}{16} \times \frac{\sqrt{7}}{4} \\ = \frac{27}{64} + \frac{35}{64} = 31/32 \end{aligned}$$

Hence, $c = 6$ satisfy the equation

Question4

In a $\triangle ABC$, if $A = 30^\circ$ and $\frac{b}{(\sqrt{3}+1)^2+2(\sqrt{2}-1)} = \frac{c}{(\sqrt{3}+1)^2-2(\sqrt{2}-1)}$, then B

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Options:

A.

60°

B.

97.5°

C.

75°

D.

52.5°

Answer: B

Solution:

Given,

$$\begin{aligned} A &= 30^\circ \\ \frac{b}{(\sqrt{3}+1)^2 + 2(\sqrt{2}-1)} &= \frac{c}{(\sqrt{3}+1)^2 - 2(\sqrt{2}-1)} \\ \Rightarrow \frac{b}{c} &= \frac{(\sqrt{3}+1)^2 + 2(\sqrt{2}-1)}{(\sqrt{3}+1)^2 - 2(\sqrt{2}-1)} \end{aligned}$$

By compounds and dividendo

$$\frac{b+c}{b-c} = \frac{2(\sqrt{3}+1)^2}{4(\sqrt{2}-1)} \Rightarrow \frac{b+c}{b-c} = \frac{2+\sqrt{3}}{\sqrt{2}-1}$$

By compounds and dividendo

$$\begin{aligned} \frac{2b}{2c} &= \frac{2 + \sqrt{3} + \sqrt{2} - 1}{2 + \sqrt{3} - \sqrt{2} + 1} \\ \Rightarrow \frac{b}{c} &= \frac{\sqrt{3} + \sqrt{2} + 1}{\sqrt{3} - \sqrt{2} + 3} \Rightarrow \frac{\sin B}{\sin C} = \frac{\sqrt{3} + \sqrt{2} + 1}{\sqrt{3} - \sqrt{2} + 3} \\ \Rightarrow \frac{\sin C}{\sin B} &= \frac{3 + \sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2} + 1} \\ \Rightarrow \frac{\sin(150^\circ - B)}{\sin B} &= \frac{3 + \sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2} + 1} \\ \{ \because A = 30^\circ \Rightarrow B + C = 150^\circ \} \\ \Rightarrow \frac{1}{2} \cdot \frac{\cos B}{\sin B} + \frac{\sqrt{3} \sin B}{2} \cdot \frac{3 + \sqrt{3} - \sqrt{2}}{\sin B} &= \frac{\sqrt{3} + \sqrt{2} + 1}{2} \\ \Rightarrow \frac{\cot B}{2} &= \frac{3 + \sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2} + 1} - \frac{\sqrt{3}}{2} \\ \Rightarrow \frac{\cot B}{2} &= \frac{6 + 2\sqrt{3} - 2\sqrt{2} - 3 - \sqrt{6} - \sqrt{3}}{2(\sqrt{3} + \sqrt{2} + 1)} \\ \cot B &= \frac{3 + \sqrt{3} - 2\sqrt{2} - \sqrt{6}}{\sqrt{3} + \sqrt{2} + 1} < 0 \\ \therefore \cot B &= \text{Negative} \\ \therefore B &> 90^\circ \end{aligned}$$

Question5

In $\triangle ABC$ is the line joining the circumcentre and the incentre is parallel to BC , then $\cos B + \cos C =$

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Options:

A.

$\frac{1}{2}$

B.

$\frac{3}{4}$

C.

1

D.

$\frac{3}{2}$

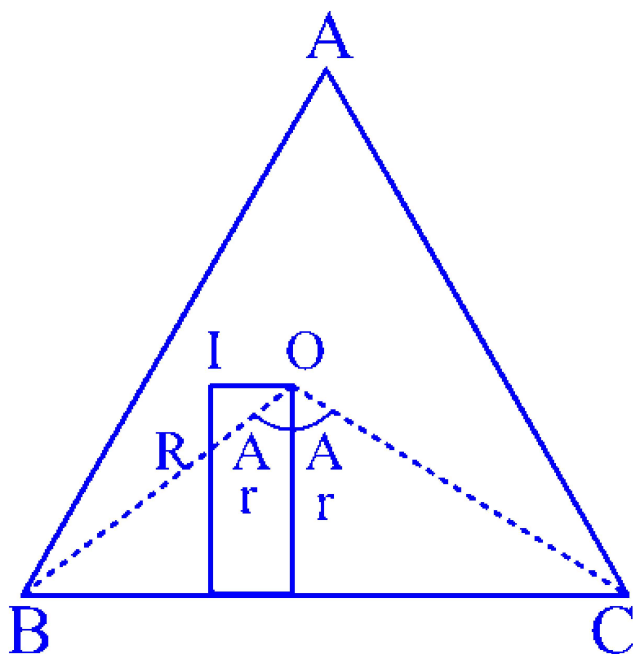
Answer: C

Solution:

O-circumcentre

I = Incentre

$$\cos A = \frac{r}{R}$$



$$\therefore \cos A + \cos B + \cos C$$

$$= 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\begin{aligned}
&= 1 + 4 \cdot \frac{r}{4R} \left(\because r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right) \\
&= 1 + \frac{r}{R} \\
\therefore \frac{r}{R} + \cos B + \cos C &= 1 + \frac{r}{R} \\
\Rightarrow \cos B + \cos C &= 1
\end{aligned}$$

Question6

In a $\triangle ABC$, if $r_1 : r_2 = 3 : 4$ and $r_2 : r_3 = 2 : 3$, then $a : b : c =$

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Options:

A.

$2 : 3 : 4$

B.

$3 : 4 : 5$

C.

$4 : 5 : 6$

D.

$5 : 6 : 7$

Answer: D

Solution:

Let's first find the common ratio $r_1 : r_2 : r_3$.

We are given:

$$1. r_1 : r_2 = 3 : 4$$

$$2. r_2 : r_3 = 2 : 3$$

To combine these ratios, we need to make the value corresponding to r_2 common.

We can write the second ratio as:

$$r_2 : r_3 = 2 : 3 = (2 \times 2) : (3 \times 2) = 4 : 6.$$

Now, we have:

$$r_1 : r_2 = 3 : 4$$

$$r_2 : r_3 = 4 : 6$$

Combining them gives the ratio $r_1 : r_2 : r_3 = 3 : 4 : 6$.

Next, we use the formulas for the exradii r_1, r_2, r_3 in terms of the area of the triangle Δ and the semi-perimeter $s = \frac{a+b+c}{2}$:

$$r_1 = \frac{\Delta}{s-a}$$

$$r_2 = \frac{\Delta}{s-b}$$

$$r_3 = \frac{\Delta}{s-c}$$

So, the ratio $r_1 : r_2 : r_3$ can be written as:

$$\frac{\Delta}{s-a} : \frac{\Delta}{s-b} : \frac{\Delta}{s-c} = 3 : 4 : 6.$$

Dividing by Δ (since $\Delta \neq 0$ for a triangle), we get:

$$\frac{1}{s-a} : \frac{1}{s-b} : \frac{1}{s-c} = 3 : 4 : 6.$$

This implies that there exists some constant k (let's use K to avoid confusion with k for side ratios) such that:

$$\frac{1}{s-a} = 3K \implies s-a = \frac{1}{3K}$$

$$\frac{1}{s-b} = 4K \implies s-b = \frac{1}{4K}$$

$$\frac{1}{s-c} = 6K \implies s-c = \frac{1}{6K}$$

Let's denote $1/K$ as C for simplicity. Then:

$$s-a = C/3$$

$$s-b = C/4$$

$$s-c = C/6$$

We know the relationships between the sides a, b, c and $s-a, s-b, s-c$:

$$a = (s-b) + (s-c)$$

$$b = (s-a) + (s-c)$$

$$c = (s-a) + (s-b)$$

Substituting the expressions for $s-a, s-b, s-c$:

$$a = \frac{C}{4} + \frac{C}{6} = \frac{3C+2C}{12} = \frac{5C}{12}$$

$$b = \frac{C}{3} + \frac{C}{6} = \frac{2C+C}{6} = \frac{3C}{6} = \frac{C}{2}$$

$$c = \frac{C}{3} + \frac{C}{4} = \frac{4C+3C}{12} = \frac{7C}{12}$$

Now we can find the ratio $a : b : c$:

$$a : b : c = \frac{5C}{12} : \frac{C}{2} : \frac{7C}{12}$$

To simplify the ratio, we can multiply all parts by $12/C$:

$$a : b : c = \left(\frac{5C}{12} \times \frac{12}{C}\right) : \left(\frac{C}{2} \times \frac{12}{C}\right) : \left(\frac{7C}{12} \times \frac{12}{C}\right)$$

$$a : b : c = 5 : 6 : 7$$

The final answer is $\boxed{5 : 6 : 7}$.

Question7

In a $\triangle ABC$, if a, b, c are in arithmetic progression and the angle A is twice the angle C , then $\cos A : \cos B : \cos C =$

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Options:

A.

$$2 : 3 : 4$$

B.

$$3 : 4 : 8$$

C.

$$2 : 9 : 12$$

D.

$$1 : 9 : 6$$

Answer: C

Solution:

Since, a, b and c are in AP

$$\text{So, } 2b = a + c$$

$$\text{Also, } \angle A = 2\angle C \quad \dots (i)$$

Now, in a $\triangle ABC$,

$$a = 2R \sin A, b = 2R \sin B, c = 2R \sin C$$

$$\text{So, } 2b = a + c$$

$$\Rightarrow 2 \cdot 2R \sin B = 2R \sin A + 2R \sin C$$

$$\Rightarrow 2 \sin B = \sin A + \sin C$$

In a $\triangle ABC$,

$$A + B + C = 180^\circ$$

$$\Rightarrow 2C + B + C = 180^\circ$$

$$\Rightarrow B = 180^\circ - 3C$$

$$\begin{aligned} \text{So, } \sin B &= \sin(180^\circ - 3C) \\ &= \sin(3C) \end{aligned} \quad \dots (ii)$$

$$\text{And } 2 \sin B = \sin A + \sin C$$

$$\Rightarrow 2 \sin(3C) = \sin(2C) + \sin C$$

[using Eq. (i) and (ii)]

$$= 2 \sin C \cos C + \sin C$$

$$\text{and } \sin(3C) = 3 \sin C - 4 \sin^3 C$$

$$\begin{aligned} \text{So, } 2 \sin(3C) &= 2(3 - 4 \sin^2 C) \\ &= 2 \cos C + 1 \\ &= 2(4 \cos^2 C - 1) \end{aligned}$$

$$\Rightarrow 8 \cos^2 C - 2 = 2 \cos C + 1$$

$$\Rightarrow 8 \cos^2 C - 2 \cos C - 3 = 0$$

$$\begin{aligned} \therefore \cos C &= \frac{2 \pm \sqrt{4 - 4(8)(-3)}}{16} \\ &= \frac{2 \pm \sqrt{100}}{16} = \frac{2 \pm 10}{16} \end{aligned}$$

$$\text{So, } \cos C = \frac{3}{4}, \cos C = -\frac{1}{2}$$

Since, C must be positive,

$$\text{So, } \cos C = \frac{3}{4}$$

$$\text{Now, } \cos A = \cos(2C)$$

$$= 2 \cos^2 C - 1 = 2\left(\frac{3}{4}\right)^2 - 1 = \frac{1}{8}$$

$$\text{And } B = 180^\circ - 3C$$

$$\begin{aligned} \cos B &= \cos(180^\circ - 3C) \\ &= -\cos(3C) \\ &= -(4 \cos^3 C - 3 \cos C) \end{aligned}$$

$$\begin{aligned}
 &= - \left(4 \left(\frac{27}{64} \right) - 3 \cdot \frac{3}{4} \right) \\
 &= - \left(\frac{27}{16} - \frac{9}{4} \right) \\
 &= - \left(\frac{27 - 36}{16} \right) = \frac{9}{16}
 \end{aligned}$$

$$\therefore \cos B = \frac{9}{16}$$

So, $\cos A : \cos B : \cos C$

$$\begin{aligned}
 &= \frac{1}{8} : \frac{9}{16} : \frac{3}{4} \\
 &= \frac{2 : 9 : 12}{16}
 \end{aligned}$$

So, $\cos A : \cos B : \cos C = 2 : 9 : 12$

Question8

In a $\triangle ABC$, A , B and C are in arithmetic progression, $rr_3 = r_1 r_2$ and $c = 10$, then $a^2 + b^2 + c^2 =$

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Options:

A.

128

B.

392

C.

288

D.

200

Answer: D

Solution:

In a triangle A, B and C are in arithmetic progression.

$$\text{So, } B = \frac{\pi}{3}$$

$$\text{Now, } rr_3 = r_1 r_2$$

$$\begin{aligned} \Rightarrow \frac{\Delta}{s} \cdot \frac{\Delta}{s-c} &= \frac{\Delta}{s-a} \cdot \frac{\Delta}{s-b} \\ \Rightarrow \frac{1}{s(s-c)} &= \frac{1}{(s-a)(s-b)} \\ \Rightarrow s^2 - (a+b)s + ab &= s^2 - cs \\ \Rightarrow ab &= (a+b-c)s \\ \Rightarrow ab &= (a+b-c) \frac{(a+b+c)}{2} \\ \Rightarrow 2ab &= (a+b)^2 - c^2 \\ \Rightarrow 2ab &= a^2 + b^2 + 2ab - c^2 \\ \Rightarrow a^2 + b^2 &= c^2 \quad \dots (i) \end{aligned}$$

Also, using cosine rule,

$$\begin{aligned} b^2 &= a^2 + c^2 - 2ac \cos B \\ \Rightarrow b^2 &= a^2 + c^2 - 2ac \cos \left(\frac{\pi}{3} \right) \\ \Rightarrow b^2 &= a^2 + c^2 - ac \\ \Rightarrow c^2 - a^2 &= a^2 + c^2 - ac \\ \Rightarrow 2a^2 &= ac \Rightarrow 2a = c \\ \Rightarrow a &= \frac{c}{2} = \frac{10}{2} = 5 \end{aligned}$$

$$\text{So, } b^2 = c^2 - a^2 = 100 - 25 = 75$$

$$\Rightarrow b = 5\sqrt{3}$$

$$\text{Now, } a^2 + b^2 + c^2 = 25 + 75 + 100 = 200$$

Question9

$$\text{In a } \triangle ABC, \frac{2(r_1+r_3)}{ac(1+\cos B)} =$$

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Options:

A.

$$\frac{\Delta}{b}$$

B.

$$\frac{b}{\Delta}$$

C.

$$\frac{2\Delta}{2+b+c}$$

D.

$$\frac{a+b+c}{2\Delta}$$

Answer: B

Solution:

$$\begin{aligned}\text{LHS} &= \frac{2(r_1+r_3)}{ac(1+\cos B)} \\ &= \frac{2(r_1+r_3)}{ac\left(1+2\cos^2\frac{B}{2}-1\right)} \\ &= \frac{2(r_1+r_3)}{ac\left(2\cos^2\frac{B}{2}\right)}\end{aligned}$$

We know that, $r_1 = \frac{\Delta}{s-a}$, $r_3 = \frac{\Delta}{s-c}$

$$\text{And } \cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$$

$$\begin{aligned}\text{Hence, LHS} &= \frac{2\left[\frac{\Delta}{s-a} + \frac{\Delta}{s-c}\right]}{ac \cdot 2\left(\sqrt{\frac{s(s-b)}{ac}}\right)^2} \\ &= \frac{2\Delta\left[\frac{1}{s-a} + \frac{1}{s-c}\right]}{2ac \cdot \left(\frac{s(s-b)}{ac}\right)} \\ &= \frac{\Delta\left[\frac{s-c+s-a}{(s-a)(s-c)}\right]}{s(s-b)} \\ &= \frac{\Delta(2s-a-c)}{s(s-a)(s-b)(s-c)} \\ &= \frac{\Delta(a+b+c-a-c)}{s(s-a)(s-b)(s-c)} \\ &= \frac{b\Delta}{s(s-a)(s-b)(s-c)} \\ &= \frac{b}{\Delta} = \text{RHS}\end{aligned}$$

Question10

In $\triangle ABC$, if $a = 8, b = 10, c = 12$, then $\frac{r}{R} =$

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Options:

A.

$$\frac{8}{15}$$

B.

$$\frac{7}{16}$$

C.

$$\frac{3}{5}$$

D.

$$\frac{5}{8}$$

Answer: B

Solution:

$$\because a = 8, b = 10 \text{ and } c = 12$$

$$\text{and } r = \frac{\Delta}{s}, R = \frac{abc}{4\Delta}$$

$$\text{So, } \frac{r}{R} = \frac{4\Delta^2}{s \cdot abc}$$

$$\because s = \frac{a+b+c}{2} = 15$$

$$\text{also, } \Delta = \sqrt{15(15-8)(15-10)(15-12)}$$

$$= \sqrt{15 \times 7 \times 5 \times 3} = 15\sqrt{7}$$

$$\text{Thus, } \frac{r}{R} = \frac{4 \times 225 \times 7}{15 \times 8 \times 10 \times 12} = \frac{7}{16}$$

Question11

In $\triangle ABC$, if $a = 13, b = 8, c = 7$, then $\cos(B + C) =$

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Options:

A. $\frac{11}{13}$

B. $\frac{23}{26}$

C. $\frac{3}{4}$

D. $\frac{1}{2}$

Answer: D

Solution:

In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$

So, $\angle B + \angle C = 180^\circ - \angle A$

$$\Rightarrow \cos(B + C) = \cos(180^\circ - A) \\ = -\cos A$$

$$\begin{aligned} \text{Now, } \cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{8^2 + 7^2 - 13^2}{2 \times 8 \times 7} \\ &= \frac{64 + 49 - 169}{112} = -\frac{56}{112} \\ &= -\frac{1}{2} \end{aligned}$$

Hence, $\cos(B + C) = -\left(-\frac{1}{2}\right) = \frac{1}{2}$

Question12

In a $\triangle ABC$, if $(r_1 - r_3)(r_1 - r_2) - 2r_2r_3 = 0$, then $a^2 - b^2 =$

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Options:

A.

$$c^2 + \frac{b^2}{4}$$

B.

$$c^2$$

C.

$$abc$$

D.

$$\frac{b+a}{c}$$

Answer: B

Solution:

$$\because (r_1 - r_3)(r_1 - r_2) - 2r_2r_3 = 0$$

and we know that

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c}$$

substitute these values

$$\Rightarrow \Delta^2 \left(\frac{1}{s-a} - \frac{1}{s-c} \right) \left(\frac{1}{s-a} - \frac{1}{s-b} \right) - 2\Delta^2 \left(\frac{1}{(s-b)(s-c)} \right) = 0$$

$$\Rightarrow \left(\frac{s-c-s+a}{(s-a)(s-c)} \right) \left(\frac{s-b-s+a}{(s-a)(s-b)} \right) - \left(\frac{2}{(s-b)(s-c)} \right) = 0$$

$$\Rightarrow \frac{1}{(s-b)(s-c)} \left[\frac{(a-c)(a-b)}{(s-a)^2} - 2 \right] = 0$$

$$\Rightarrow (a-c)(a-b) - 2(s-a)^2 = 0$$

$$\Rightarrow (a-c)(a-b) - 2 \left(\frac{a+b+c}{2} - a \right)^2 = 0$$

$$\Rightarrow 2(a-c)(a-b) - (b+c-a)^2 = 0$$

$$\Rightarrow 2(a^2 - ab - ac + bc) - (b^2 + c^2 + a^2) - 2(bc - ca - ab) = 0$$

$$\Rightarrow a^2 - b^2 - c^2 - 2ab - 2ac + 2bc - 2bc + 2ac + 2ab = 0$$

$$\Rightarrow a^2 - b^2 = c^2$$

Question13

If the median AD of the $\triangle ABC$ is bisected at E and BE meets AC in F , then $AF : AC =$

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Options:

A.

1 : 4

B.

1 : 3

C.

1 : 2

D.

3 : 4

Answer: B

Solution:

$\therefore D$ is the mid-point of BC and E is the mid-point of AD .

Let BE and AC intersect at F

So, $\frac{AF}{FC} = \frac{1}{2}$

and $AC = AF + FC = \frac{1}{3}AC + \frac{2}{3}AC$

Thus, $\frac{AF}{AC} = \frac{\frac{1}{3}AC}{AC} = \frac{1}{3}$

Hence, $AF : AC = 1 : 3$

Question14

In $\triangle ABC$ if $\cos A \cos B + \sin A \sin B \sin C = 1$, then $\sin A + \sin B + \sin C =$

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Options:

A.

$$\frac{2+\sqrt{3}}{2}$$

B.

$$1 + \sqrt{2}$$

C.

$$\frac{2\sqrt{3}-1}{2}$$

D.

$$\frac{3+\sqrt{3}}{2}$$

Answer: B

Solution:

We have,

$$\cos A \cos B + \sin A \cdot \sin B \sin C = 1$$

In $\triangle ABC$,

$\cos A \cos B + \sin A \sin B \sin C = 1$ is possible if $A = 45^\circ$, $B = 45^\circ$ and $C = 90^\circ$ i.e.

$$\cos 45^\circ \cdot \cos 45^\circ + \sin 45^\circ \cdot \sin 45^\circ \cdot \sin 90^\circ$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

$$\therefore \sin A + \sin B + \sin C$$

$$= \sin 45^\circ + \sin 45^\circ + \sin 90^\circ$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 1 = \sqrt{2} + 1$$

Question15

In $\triangle ABC$, if $a : b : c = 4 : 5 : 6$, then $\frac{\cos A + 3 \cos C}{\cos B} =$

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Options:

A.

1

B.

4

C.

2

D.

3

Answer: C

Solution:

Given, $a : b : c = 4 : 5 : 6$

We can express the sides as $a = 4x$, $b = 5x$ and $c = 6x$ for some positive constant x .

$$\begin{aligned}\therefore \cos A &= \frac{b^2 + c^2 - a^2}{2bc} \text{ (By Cosine rule)} \\ &= \frac{(5x)^2 + (6x)^2 - (4x)^2}{2 \times 5x \times 6x} = \frac{45x^2}{60x^2} = \frac{3}{4}\end{aligned}$$

$$\begin{aligned}\text{Similarly, } \cos B &= \frac{(4x)^2 + (6x)^2 - (5x)^2}{2(4x)(6x)} \\ &= \frac{27x^2}{48x^2} = \frac{9}{16}\end{aligned}$$

$$\begin{aligned}\text{and } \cos C &= \frac{(4x)^2 + (5x)^2 - (6x)^2}{2(4x)(5x)} \\ &= \frac{5x^2}{40x^2} = \frac{1}{8}\end{aligned}$$

$$\begin{aligned}\text{So, } \frac{\cos A + 3 \cos C}{\cos B} &= \frac{\frac{3}{4} + 3\left(\frac{1}{8}\right)}{\frac{9}{16}} = \frac{\frac{6+3}{8}}{\frac{9}{16}} \\ &= \frac{9}{8} \times \frac{16}{9} = 2\end{aligned}$$

Question16

In $\triangle ABC$, if $a = 6$, $b = 8$ and $c = 10$, then $\frac{2r_2r_3}{rr_1} =$

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Options:

A.

$$b + c$$

B.

$$c + a$$

C.

$$a + b$$

D.

$$a + b + c$$

Answer: A

Solution:

Given, $a = 6, b = 8, c = 10$

$$\therefore s = \frac{a + b + c}{2} = 12$$

$$\begin{aligned} \text{and area } (\Delta) &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{12 \times 6 \times 4 \times 2} = 24 \end{aligned}$$

Now, $r_1 = \frac{\Delta}{s-a} = \frac{24}{6} = 4$ and

$$r = \frac{\Delta}{s} = \frac{24}{12} = 2$$

$$r_2 = \frac{\Delta}{s-b} = \frac{24}{4} = 6$$

$$r_3 = \frac{\Delta}{s-c} = \frac{24}{2} = 12$$

$$\begin{aligned} \therefore \frac{2r_2r_3}{r_1} &= \frac{2 \times 6 \times 12}{2 \times 4} = 18 \\ &= 8 + 10 = b + c \end{aligned}$$

Question17

If the sides a, b, c of the $\triangle ABC$ are in harmonic progression, then $\operatorname{cosec}^2 A/2, \operatorname{cosec}^2 B/2, \operatorname{cosec}^2 C/2$ are in

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Options:

A.

Arithmetico-geometric progression

B.

Arithmetic progression

C.

Geometric progression

D.

Harmonic progression

Answer: B

Solution:

a, b, c are in HP.

$$\Rightarrow b = \frac{2ac}{a+c}$$

Let $\operatorname{cosec}^2 \frac{A}{2}, \operatorname{cosec}^2 \frac{B}{2}, \operatorname{cosec}^2 \frac{C}{2}$ are in AP .

$$\Rightarrow \frac{1}{\sin^2 \frac{A}{2}}, \frac{1}{\sin^2 \frac{B}{2}}, \frac{1}{\sin^2 \frac{C}{2}} \text{ are in AP.}$$

$$\Rightarrow \frac{bc}{(s-b)(s-c)}, \frac{ac}{(s-a)(s-c)}$$

$$\frac{ab}{(s-a)(s-b)} \text{ are in AP.}$$

$$\Rightarrow \frac{bc}{s(s-b)(s-c)}, \frac{ac}{s(s-a)(s-c)}$$

$$\frac{ab}{s(s-a)(s-b)} \text{ are in AP.}$$

$$\Rightarrow \frac{bc(s-a)}{s(s-a)(s-b)(s-c)}, \frac{ac(s-b)}{s(s-a)(s-b)(s-c)}, \frac{ab(s-c)}{s(s-a)(s-b)(s-c)} \text{ are in AP.}$$

$$\Rightarrow bcs - abc, acs - abc, abs - abc \text{ are in AP.}$$

$$\Rightarrow bcs, acs, abs \text{ are in AP.}$$

$$\Rightarrow bc, ac, ab \text{ are in AP.}$$

$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP.

$\Rightarrow a, b, c$ are in HP. Which is true.

Question18

In $\triangle ABC$, if $r = 3$ and $R = 5$, then $\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} =$

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Options:

A.

$$\frac{1}{30}$$

B.

$$\frac{12}{15}$$

C.

$$\frac{1}{15}$$

D.

$$\frac{5}{36}$$

Answer: A

Solution:

$$r = 3, R = 5$$

$$\begin{aligned} \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} &= \frac{a+b+c}{abc} = \frac{2s}{abc} \\ &= \frac{2s}{abc} \times \frac{\Delta}{\Delta} \\ &= \frac{1}{2} \cdot \frac{4\Delta}{abc} \cdot \frac{s}{\Delta} \\ &= \frac{1}{2} \cdot \frac{1}{R} \cdot \frac{1}{r} = \frac{1}{2} \cdot \frac{1}{5} \cdot \frac{1}{3} = \frac{1}{30} \end{aligned}$$

Question19

In a $\triangle ABC$, $A - B = 120^\circ$, $R = 8r$, then $\frac{1+\cos C}{1-\cos C} =$

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Options:

A.

16

B.

14

C.

15

D.

10

Answer: C

Solution:

Given in $\triangle ABC$,

$$A - B = 120^\circ, R = 8r$$

$$\frac{1 + \cos C}{1 - \cos C}$$

$$\Rightarrow R = 8 \left(4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \right)$$

$$\Rightarrow 2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{1}{16}$$

$$\Rightarrow \left(\cos \frac{A-B}{2} - \cos \left(\frac{A+B}{2} \right) \right) \sin \frac{C}{2} = \frac{1}{16}$$

$$\Rightarrow \sin \frac{C}{2} \left(\frac{1}{2} - \sin \frac{C}{2} \right) = \frac{1}{16}$$

$$\sin \frac{C}{2} = \frac{1}{4}$$

$$\cos C = 1 - \frac{2}{16} = \frac{7}{8}$$

$$\Rightarrow \frac{1 + \cos C}{1 - \cos C} = 15$$

Question20

In $\triangle ABC$, $\sqrt{\frac{r \cdot r_2}{r_3 r_1}} =$

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Options:

A.

$$(r_3 - r_2)(r_1 - r_2)$$

B.

$$r_3 + r_1$$

C.

$$\frac{b}{r_3 - r_1}$$

D.

$$\frac{b}{r_3 + r_1}$$

Answer: D

Solution:

Given, $\sqrt{\frac{r_2}{r_3 r_1}}$, in $\triangle ABC$

$$\begin{aligned}\sqrt{\frac{r_2}{r_3 r_1}} &= \sqrt{\frac{\frac{\Delta}{s} - \frac{\Delta}{s-b}}{\frac{\Delta}{s-c} \frac{\Delta}{s-a}}} \sqrt{\frac{\frac{\Delta}{s} \frac{\Delta}{s-b}}{\frac{\Delta}{s-c} \frac{\Delta}{s-a}}} \\ &= \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} \\ \Rightarrow s &= \frac{a+b+c}{2} \\ \Rightarrow 2s &= a+b+c \Rightarrow 2s-a-c = b\end{aligned}$$

$$\begin{aligned}\text{Now, } \frac{b}{r_3 + r_1} &= \frac{b}{\frac{\Delta}{s-c} + \frac{\Delta}{s-a}} \\ &= \frac{b}{\frac{\Delta(2s-a-c)}{(s-a)(s-c)}} = \frac{b(s-a)(s-c)}{\Delta b} \\ &= \frac{(s-a)(s-c)}{\sqrt{s(s-a)(s-b)(s-c)}} \\ &= \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} = \sqrt{\frac{r r_2}{r_3 r_1}}\end{aligned}$$

Question21

If $A(0, 0, 0)$, $B(3, 4, 0)$ and $C(0, 12, 5)$ are the vertices of a $\triangle ABC$, then the x -coordinate of its incentre is

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Options:

A.

$$\frac{25}{18+7\sqrt{2}}$$

B.

$$\frac{25}{26}$$

C.

$$\frac{39}{18+7\sqrt{2}}$$

D.

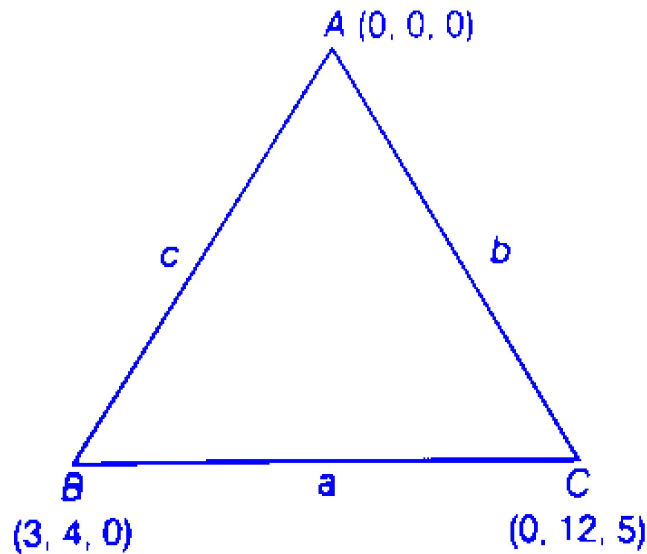
$$\frac{39}{26}$$

Answer: C

Solution:

x -coordinate of incentre

$$= \frac{ax_1 + by_1 + cz_1}{a + b + c}$$



$$a = \sqrt{98}$$

$$b = 13$$

$$c = 5$$

$$\begin{aligned} I_x &= \frac{\sqrt{98} \times 0 + 13 \times 3 + 5 \times 0}{18 + \sqrt{98}} \\ &= \frac{39}{18 + \sqrt{98}} = \frac{39}{18 + 7\sqrt{2}} \end{aligned}$$

Question22

In a $\triangle ABC$, if $\sin \frac{A}{2} = \frac{1}{4} \sqrt{\frac{3}{5}}$, $a = 2$, $c = 5$ and b is an integer, then the area (in sq. units) of $\triangle ABC$ is

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Options:

A.

$$\frac{\sqrt{297}}{4}$$

B.

$$\frac{\sqrt{231}}{4}$$

C.

$$\frac{\sqrt{385}}{4}$$

D.

$$\frac{\sqrt{185}}{4}$$

Answer: B

Solution:

$$\sin \frac{A}{2} = \frac{1}{4} \sqrt{\frac{3}{5}} a = 2c = 5$$

$$\cos A = 1 - 2 \sin^2 \frac{A}{2} = 1 - 2 \cdot \frac{1}{16} \times \frac{3}{5} = \frac{37}{40}$$

$$\text{Now, } \cos A = \frac{b^2 + c^2 - a^2}{2 \times b \times c}$$

$$\Rightarrow \frac{37}{40} = \frac{b^2 + 25 - 4}{2 \times b \times 5}$$

$$\Rightarrow \frac{37}{4} = \frac{b^2 + 21}{b}$$

$$\Rightarrow 4b^2 - 37b + 84 = 0$$

$$\therefore (4b - 21)(b - 4) = 0$$

$\therefore b = 4$ because given b is an integer

$$\therefore \sin A = \sqrt{1 - \cos^2 A} = \sqrt{1 - \left(\frac{37}{40}\right)^2}$$

$$\therefore \text{Area of triangle} = \frac{1}{2}b \times c \times \sin A$$

$$= \frac{1}{2} \times 4 \times 5 \sqrt{1 - \left(\frac{37}{40}\right)^2}$$

$$= 10 \sqrt{\frac{77}{40} \times \frac{3}{40}} = \frac{1}{4} \sqrt{231}$$

Question23

In a $\triangle ABC$ if $a + c = 5b$, then $\cot \frac{A}{2} \cot \frac{C}{2} =$

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Options:

A.

2

B.

$\frac{1}{2}$

C.

$\frac{3}{2}$

D.

$\frac{2}{3}$

Answer: C

Solution:

$$\begin{aligned}\text{Given, } a + c &= 5b \\ a + b + c &= 2s = 6b \\ s &= 3b\end{aligned}$$

$$\begin{aligned}\cot \frac{A}{2} \cdot \cot \frac{C}{2} &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \\ &\times \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\ &= \frac{s}{s-b} = \frac{3b}{3b-b} = \frac{3b}{2b} = \frac{3}{2}\end{aligned}$$

Question24

In a $\triangle ABC$, if $r_1 = 3, r_2 = 4, r_3 = 6$, then $b =$

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Options:

A.

$$2\sqrt{6}$$

B.

$$\frac{5\sqrt{6}}{3}$$

C.

$$\frac{7\sqrt{6}}{3}$$

D.

$$3\sqrt{6}$$

Answer: A

Solution:

$$\text{Given, } r_1 = 3, r_2 = 4 \text{ and } r_3 = 6$$

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c}$$

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{s}{\Delta}$$

$$\frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{s}{\Delta} \Rightarrow \frac{s}{\Delta} = \frac{3}{4}$$

And $r_1 r_2 + r_2 r_3 + r_1 r_3 = s^2$

$$\Rightarrow 3 \times 4 + 4 \times 6 + 3 \times 6 = s^2$$

$$\Rightarrow 12 + 24 + 18 = s^2$$

$$\Rightarrow s^2 = 54$$

$$\Rightarrow s = 3\sqrt{6}$$

$$\therefore \Delta = \frac{4s}{3} = \frac{12\sqrt{6}}{3} = 4\sqrt{6}$$

$$r_2 = \frac{\Delta}{s-b} \Rightarrow s-b = \frac{\Delta}{r_2}$$

$$b = s - \frac{\Delta}{r_2} = 3\sqrt{6} - \frac{4\sqrt{6}}{4}$$

$$= 3\sqrt{6} - \sqrt{6} = 2\sqrt{6}$$

Question25

In $\triangle ABC$, the sum of the lengths of two sides is x and the product of those lengths is y . If c is the length of its third side and $x^2 - c^2 = y$, then the circumradius of that triangle is

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Options:

A.

$$\frac{c}{\sqrt{3}}$$

B.

$$\frac{c}{3}$$

C.

$$\frac{y}{\sqrt{3}}$$

D.

$$\frac{3y}{2}$$

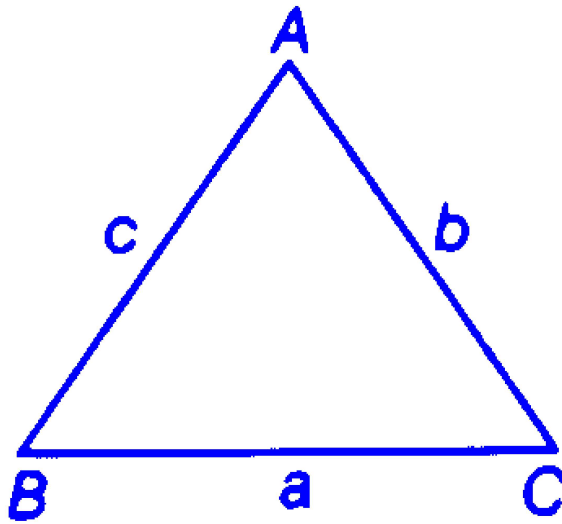
Answer: A

Solution:

We have,

$$x = a + b$$

$$y = ab$$



We have, $x^2 - c^2 = y$

$$\Rightarrow (a + b)^2 - c^2 = ab$$

$$\Rightarrow a^2 + b^2 - c^2 = -ab$$

$$\Rightarrow \frac{a^2 + b^2 - c^2}{2ab} = -\frac{1}{2}$$

$$\cos c = -\frac{1}{2} \Rightarrow c = \frac{2\pi}{3} \Rightarrow \sin c = \frac{\sqrt{3}}{2}$$

Using circumradius formula,

$$c = 2R \sin c$$

$$R = \frac{c}{\frac{2\sqrt{3}}{2}} = \frac{c}{\sqrt{3}}$$

Question26

If the area of a $\triangle ABC$ is $4\sqrt{5}$ sq units. Length of the side CA is 6 units and $\tan \frac{B}{2} = \frac{\sqrt{5}}{4}$, then its smallest side is of length

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Options:

A.

5 units

B.

4 units

C.

3 units

D.

6 units

Answer: C

Solution:

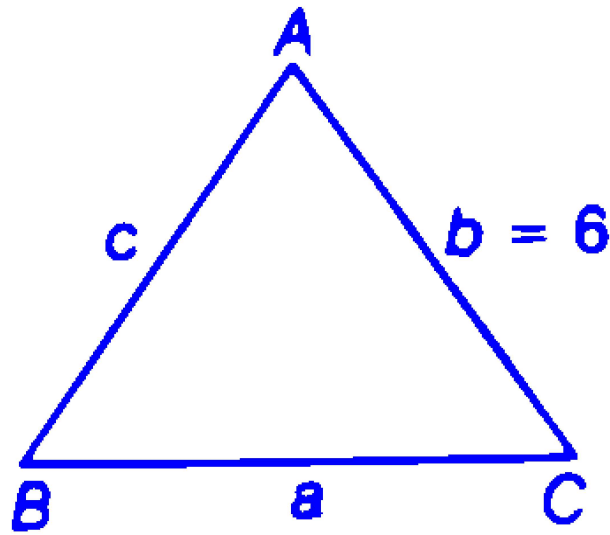
We have, $\tan \frac{B}{2} = \frac{\sqrt{5}}{4}$ and $b = 6$

Area of $\triangle ABC = \Delta = 4\sqrt{5}$

$$\cos B = \frac{1 - \tan^2 \frac{B}{2}}{1 + \tan^2 \frac{B}{2}} = \frac{11}{21}$$

$$\Rightarrow \sin B = \frac{\sqrt{320}}{21} = \frac{8\sqrt{5}}{21}$$

$$\begin{aligned} \therefore \Delta &= \frac{1}{2}ac \sin B = 4\sqrt{5} \\ &= \frac{1}{2}ac \frac{8\sqrt{5}}{21} = 4\sqrt{5} \end{aligned}$$



$$\therefore ac = 21 \quad \dots (i)$$

$$\therefore b^2 = c^2 + a^2 - 2ac \cos B$$

$$\Rightarrow 36 = c^2 + a^2 - 2 \times 21 \times \frac{11}{21}$$

$$\Rightarrow 36 = c^2 + a^2 - 22$$

$$\Rightarrow c^2 + a^2 = 58$$

$$\therefore (c + a)^2 = c^2 + a^2 + 2ac \\ = 58 + 42 = 100$$

$$c + a = 10 \quad \dots (ii)$$

$$\Rightarrow c = 7 \text{ and } a = 3$$

$\therefore$ Smallest sides is 3 units.

Question27

In a $\triangle ABC$ if $r_1 = 2r_2 = 3r_3$, then $a : b$ is

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Options:

A.

3 : 5

B.

5 : 3

C.

4 : 5

D.

5 : 4

Answer: D

Solution:

We have, $r_1 = 2r_2 = 3r_3$

$$\Rightarrow \frac{\Delta}{s-a} = \frac{2\Delta}{s-b} = \frac{3\Delta}{s-c} = \frac{1}{k} \quad (\text{say})$$

$$\therefore s-a = k$$

$$s-b = 2k$$

$$\underline{s-c = 3k}$$

$$3s - (a+b+c) = 6k$$

$$s = 6k$$

$$a = s - k = 5k$$

$$b = s - 2k = 4k$$

$$\therefore a : b = 5 : 4$$

Question28

In $\triangle ABC$, $a^2 \sin 2B + b^2 \sin 2A$ is equal to

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Options:

A. $2ab \cos A$

B. $2ab \sin A$

C. $2ab \sin C$

D. $2ab \cos C$

Answer: C

Solution:

In $\triangle ABC$, to evaluate the expression $a^2 \sin 2B + b^2 \sin 2A$, let's proceed with the following transformations:

$$\begin{aligned} a^2 \sin 2B + b^2 \sin 2A &= 2a^2 \sin B \cos B + 2b^2 \sin A \cos A \\ &= 2a(b \sin A) \cos B + 2b^2 \sin A \cos A \\ &= 2b \sin A(a \cos B + b \cos A). \end{aligned}$$

Now, applying the sine rule and angle sum identity in $\triangle ABC$:

$$a \cos B + b \cos A = c.$$

Thus, the expression simplifies to:

$$= 2b \sin A(c).$$

By the sine rule, we know that:

$$c = a \cos B + b \cos A \Rightarrow a^2 \sin 2B + b^2 \sin 2A = 2ab \sin C.$$

Therefore, the expression simplifies to $2ab \sin C$.

Question 29

In $\triangle ABC$, $\frac{r_2(r_1+r_3)}{\sqrt{r_1r_2+r_2r_3+r_3r_1}}$ is equal to

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Options:

A. a

B. b

C. c

D. s

Answer: B

Solution:

Given, In $\triangle ABC$,

$$\begin{aligned} \frac{r_2(r_1 + r_3)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} &= \frac{r_1r_2 + r_2r_3}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} \\ &= \frac{\frac{\Delta}{(s-a)} \times \frac{\Delta}{(s-b)} + \frac{\Delta}{(s-b)} \times \frac{\Delta}{(s-c)}}{\sqrt{\frac{\Delta}{s-a} \times \frac{\Delta}{s-b} + \frac{\Delta}{s-b} \times \frac{\Delta}{s-c} + \frac{\Delta}{s-c} \times \frac{\Delta}{s-a}}} \\ &= \frac{\frac{\Delta^2(s-c+s-a)}{(s-a)(s-b)(s-c)}}{\Delta \sqrt{\frac{(s-c)+(s-a)+(s-b)}{(s-a)(s-b)(s-c)}}} \\ &= \frac{\Delta(2s-a-c)}{(s-a)(s-b)(s-c)} \times \frac{\sqrt{(s-a)(s-b)(s-c)}}{\sqrt{3c-a-b-c}} \\ &= \frac{\Delta(b)}{\sqrt{(s-a)(s-b)(s-c)}\sqrt{s}} \\ &= \frac{\Delta(b)}{\sqrt{s(s-a)(s-b)(s-c)}} = b \end{aligned}$$

Question30

In $\triangle ABC$, $(r_2 + r_3) \operatorname{cosec}^2 \frac{A}{2}$ is equal to

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Options:

A. $4R$

B. $4R \cot^2 \frac{A}{2}$

C. $4R \tan^2 \frac{A}{2}$

D. $R \tan^2 \frac{A}{2}$

Answer: B

Solution:

We have, in $\triangle ABC$

$$\begin{aligned} & (r_2 + r_3) \operatorname{cosec}^2 \frac{A}{2} \\ &= \left(4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} + 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} \right) \operatorname{cosec}^2 \frac{A}{2} \\ &= 4R \cos \frac{A}{2} \left(\sin \frac{B}{2} \cos \frac{C}{2} + \cos \frac{B}{2} \sin \frac{C}{2} \right) \\ &= 4R \cos \frac{A}{2} \sin \left(\frac{B}{2} + \frac{C}{2} \right) \operatorname{cosec}^2 \frac{A}{2} \\ &= 4R \cos \frac{A}{2} \cdot \cos \frac{A}{2} \operatorname{cosec}^2 \frac{A}{2} \\ &= 4R \cot^2 \frac{A}{2} \end{aligned}$$

Question31

In a $\triangle ABC$, if $a = 13$, $b = 14$ and $c = 15$, then $r_1 =$

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Options:

A. $\frac{23}{2}$

B. $\frac{21}{2}$

C. $\frac{25}{2}$

D. $\frac{26}{2}$

Answer: B

Solution:

Given that in a $\triangle ABC$

$$a = 13, b = 14 \text{ and } c = 15$$

$$\text{we know that } r_1 = \frac{\Delta}{s-a}$$

$$\text{where, } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{and } s = \frac{a+b+c}{2}$$

$$\text{So, } s = \frac{13+14+15}{2} = \frac{42}{2} = 21$$

$$\begin{aligned}
 \text{Now, } \Delta &= \sqrt{21 \times 8 \times 7 \times 6} \\
 &= \sqrt{3 \times 7 \times 2 \times 2 \times 2 \times 7 \times 2 \times 3} \\
 &= 2 \times 2 \times 3 \times 7 = 84
 \end{aligned}$$

$$\text{So, } r_1 = \frac{84}{21-13} = \frac{84}{8} = \frac{21}{2}$$

Question32

In $\triangle ABC$ if $r : R : r_2 = 1 : 3 : 7$, then $\sin(A + C) + \sin B$ is equal to

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Options:

A. 0

B. $\sqrt{3}$

C. 1

D. 2

Answer: D

Solution:

Given that

$$r : R : r_2 = 1 : 3 : 7$$

we know that $A + B + C = 180^\circ$

$$\begin{aligned}
 \text{Now, } \sin(A + C) + \sin B &= \sin(180^\circ - B) + \sin B \\
 &= \sin B + \sin B = 2 \sin B
 \end{aligned}$$

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{\Delta}{s}$$

$$R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C} = \frac{abc}{4\Delta}$$

On simplifying above ratio, we get

$$B = 90^\circ$$

$$\therefore \sin(A + C) + \sin B = 2 \sin 90^\circ = 2 \times 1 = ?$$

Question33

In $\triangle ABC$, $(r_1 + r_2) \operatorname{cosec}^2 \frac{C}{2}$ is equal to

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Options:

A. $2R \cot^2 \frac{C}{2}$

B. $4R \tan^2 \frac{C}{2}$

C. $4R \cot^2 \frac{C}{2}$

D. $2R \tan^2 \frac{C}{2}$

Answer: C

Solution:

Given, in a $\triangle ABC$,

$$A + B + C = 180^\circ$$

Now, $(r_1 + r_2) \operatorname{cosec}^2 \frac{C}{2}$

$$= \left(4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} + 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} \right) \cdot \operatorname{cosec}^2 \frac{C}{2}$$

$$= 4R \cos \frac{C}{2} \left[\sin \frac{A}{2} \cos \frac{B}{2} + \cos \frac{A}{2} \sin \frac{B}{2} \right] \operatorname{cosec}^2 \frac{C}{2}$$

$$= 4R \cos \frac{C}{2} \left[\sin \left(\frac{A}{2} + \frac{B}{2} \right) \right] \operatorname{cosec}^2 \frac{C}{2}$$

$$\begin{aligned}
&= 4R \cos \frac{C}{2} \cdot \sin \frac{(A+B)}{2} \cdot \operatorname{cosec}^2 \frac{C}{2} \\
&= 4R \cos \frac{C}{2} \cdot \frac{1}{\sin^2 \frac{C}{2}} \sin \left(\frac{A+B}{2} \right) \\
&= 4R \cot^2 \frac{C}{2} \cdot \frac{\sin \left(\frac{A+B}{2} \right)}{\cos \frac{C}{2}} \\
&= 4R \cot^2 \frac{C}{2} \cdot \frac{\sin \left(90^\circ - \frac{C}{2} \right)}{\cos \frac{C}{2}} \\
&= 4R \cot^2 \frac{C}{2} \cdot \frac{\cos \frac{C}{2}}{\cos \frac{C}{2}} = 4R \cot^2 \frac{C}{2}
\end{aligned}$$

Question34

In a $\triangle ABC$, if A, B and C are in arithmetic progression and $\cos A + \cos B + \cos C = \frac{1+\sqrt{2}+\sqrt{3}}{2\sqrt{2}}$, then $\tan A$:

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Options:

A. $\sqrt{3}$

B. $2 + \sqrt{3}$

C. 1

D. $2 - \sqrt{3}$

Answer: B

Solution:

Given $\triangle ABC$, A, B, C in AP which means $B - C = A - B$ where $C > B > A$

$$\Rightarrow 2B = A + C$$

Sum of angles of $\triangle ABC$

$$A + B + C = 180^\circ \quad \dots (i)$$

$\therefore A, B, C$ in AP, we can assume that $c = a$,

$$B = a + d, A = a + 2d$$

From Eq. (i),

$$a + 2d + a + d + a = 180^\circ$$

$$3a + 3d = 180$$

$$a + d = \frac{180}{3} \Rightarrow B = 60^\circ$$

$$\text{Thus, } A + C = 120^\circ$$

$$\text{Given } \cos A + \cos B + \cos C = \frac{1+\sqrt{2}+\sqrt{3}}{2\sqrt{2}} \quad \dots (ii)$$

Let $C = 45^\circ$ (we need to check for angles)

$$\text{Then, } B = 60^\circ, A = 75^\circ$$

Using Eq. (ii),

$$\begin{aligned} \cos 75^\circ + \cos 60^\circ + \cos 45^\circ &= \frac{\sqrt{3}-1}{2\sqrt{2}} + \frac{1}{2} + \frac{1}{\sqrt{2}} \\ &= \frac{2 + \sqrt{2} + \sqrt{3} - 1}{2\sqrt{2}} = \frac{1 + \sqrt{2} + \sqrt{3}}{2\sqrt{2}} \end{aligned}$$

This matches the expression.

Therefore, $A = 75^\circ$ and $\tan A = \tan 75^\circ$

$$\tan A = 2 + \sqrt{3}$$

Question35

In $\triangle ABC$, if $b + c : c + a : a + b = 7 : 8 : 9$, then the smaller angle (in radians) of that triangle is

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Options:

A. $\cos^{-1} \left(\frac{4}{5} \right)$

B. $\frac{\pi}{3}$

C. $\cos^{-1} \left(\frac{3}{5} \right)$

D. $\frac{\pi}{4}$

Answer: A

Solution:

In triangle $\triangle ABC$, given the ratio of the sums of the sides as $b + c : c + a : a + b = 7 : 8 : 9$, we can express these sums using a common factor k as follows:

$$b + c = 7k$$

$$c + a = 8k$$

$$a + b = 9k$$

By adding up these equations, we have:

$$b + c + c + a + a + b = 7k + 8k + 9k = 24k$$

Simplifying, we get:

$$2(a + b + c) = 24k \Rightarrow a + b + c = 12k$$

Using this, we can solve for each side:

From $a + 7k = 12k$, it follows that $a = 5k$.

From $c + a = 8k$, substitute $a = 5k$ to find $c = 3k$.

From $a + b = 9k$, substitute $a = 5k$ to find $b = 4k$.

Thus, the side lengths are:

$$a = 5k$$

$$b = 4k$$

$$c = 3k$$

Next, we apply the cosine rule to find the angle C :

$$\cos C = \frac{b^2 + a^2 - c^2}{2ba}$$

Substituting the values, we have:

$$\begin{aligned}\cos C &= \frac{(4k)^2 + (5k)^2 - (3k)^2}{2 \times 4k \times 5k} \\ &= \frac{16k^2 + 25k^2 - 9k^2}{40k^2} \\ &= \frac{32k^2}{40k^2} = \frac{4}{5}\end{aligned}$$

Therefore, the smallest angle C in radians is:

$$C = \cos^{-1} \left(\frac{4}{5} \right)$$

Question 36

In $\triangle ABC$, if $(a + c)^2 = b^2 + 3ca$, then $\frac{a+c}{2R} =$

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Options:

A. $\frac{\sqrt{3}}{2}$

B. $\sqrt{3} \cos \left(\frac{A-C}{2} \right)$

C. $\cos \left(\frac{A-C}{2} \right)$

D. $\sin \left(\frac{A-C}{2} \right)$

Answer: B

Solution:

In the given triangle $\triangle ABC$, we are provided with the equation:

$$(a + c)^2 = b^2 + 3ac$$

Using the cosine rule for triangles:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Additionally, we know the sum of angles in a triangle is π , i.e., $A + B + C = \pi$.

Given the equation:

$$(a + c)^2 = b^2 + 3ac$$

We rearrange this equation:

$$a^2 + c^2 + 2ac = b^2 + 3ac$$

$$a^2 + c^2 - b^2 = ac \quad (i)$$

From the cosine rule:

$$a^2 + c^2 - b^2 = 2ac \cos B \quad (ii)$$

Equating Equation (i) and Equation (ii), we get:

$$ac = 2ac \cos B$$

$$1 = 2 \cos B$$

$$\cos B = \frac{1}{2}$$

Thus, $\angle B = \frac{\pi}{3}$ or 60° .

Therefore, we have:

$$\frac{a+c}{2R} = \sin B \Rightarrow \sin 60^\circ = \frac{\sqrt{3}}{2}$$

Thus:

$$\frac{a+c}{2R} = \sqrt{3} \cos\left(\frac{A-C}{2}\right)$$

Question37

In $\triangle ABC$, if A, B and C are in arithmetic progression $\Delta = \frac{\sqrt{3}}{2}$ and $r_1 r_2 = r_2 r$, then $R =$

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Options:

A. $\sqrt{3}$

B. 2

C. 1

D. $\sqrt{2}$

Answer: C

Solution:

Given that the angles $\angle A, \angle B, \angle C$ in $\triangle ABC$ are in arithmetic progression, we can express $\angle B$ as the average of $\angle A$ and $\angle C$, i.e., $B = \frac{A+C}{2}$.

We also know that the sum of angles in a triangle is 180° :

$$A + B + C = 180^\circ$$

Substituting $B = \frac{A+C}{2}$ into this equation gives:

$$A + \frac{A+C}{2} + C = 180^\circ$$

$$2A + A + C + 2C = 180^\circ \times 2$$

$$\frac{A+C}{2} = 60^\circ = B$$

Therefore, $B = 60^\circ$.

The area of $\triangle ABC$ is given as $\frac{\sqrt{3}}{2}$.

Regarding the radii of the excircles, denoted as r_1, r_2, r_3 for the sides opposite to vertices A, B, C respectively, the condition given is:

$$r_1 r_2 = r_2 r$$

For any triangle, the area can also be expressed as:

$$\frac{abc}{4R}$$

Thus, equating the given area:

$$\frac{\sqrt{3}}{2} = \frac{abc}{4R}$$

Solving for R :

$$R = \frac{abc}{2\sqrt{3}}$$

Since we have determined that $B = 60^\circ$, the circumradius R in such a triangle is given by:

$$R = \frac{b}{\sqrt{3}}$$

Calculating the circumradius directly for special triangles:

Hence, the circumradius R is 1.

Question38

If 7 and 8 are the length of two sides of a triangle and a' is the length of its smallest side. The angles of the triangle are in AP and ' a ' has two values a_1 and a_2 satisfying this condition. If $a_1 < a_2$, then $2a_1 + 3a_2 =$

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Options:

A. 15

B. 21

C. 24

D. 28

Answer: B

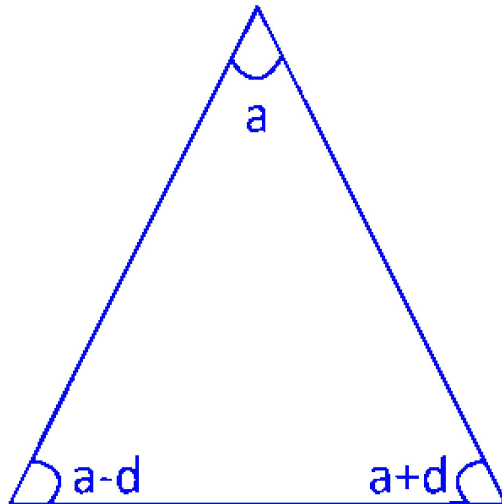
Solution:

We have, two larger side of a triangle are 8 and 7 and smaller side is a and angles are in AP

According to the given condition.

$$\Rightarrow a + a - d + a + d = 180^\circ$$

$$\Rightarrow a = 60^\circ$$



$$\text{Then, } \cos 60^\circ = \frac{a^2 + b^2 - c^2}{2ac}$$

$$\Rightarrow \frac{1}{2} = \frac{a^2 + 64 - 49}{2 \times a \times 8}$$

$$\Rightarrow a^2 - 8a + 15 = 0$$

On solving, we get

$$\Rightarrow a = \frac{8 \pm \sqrt{64 - 60}}{2} = \frac{8 \pm 2}{2}$$

$$\Rightarrow a_1 = 3, a_2 = 5$$

Hence, the value satisfy the condition a_1

$$\text{Then, } 2a_1 + 3a_2 = 2 \times 3 + 3 \times 5$$

$$\Rightarrow 2a_1 + 3a_2 = 21$$

Question39

In $\triangle ABC$, if $a = 13, b = 14$ and $\cos \frac{C}{2} = \frac{3}{\sqrt{13}}$, then $2r_1 =$

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Options:

A. $2S$

B. Δ

C. S

D. 2Δ

Answer: C

Solution:

Given the triangle $\triangle ABC$ with sides $a = 13$, $b = 14$, and $\cos\left(\frac{C}{2}\right) = \frac{3}{\sqrt{13}}$.

Step 1: Utilize the Cosine Rule

First, we find $\cos C$ using:

$$\cos C = 2 \cos^2\left(\frac{C}{2}\right) - 1$$

Substitute the given value:

$$\cos C = 2\left(\frac{3}{\sqrt{13}}\right)^2 - 1 = 2 \times \frac{9}{13} - 1 = \frac{18-13}{13} = \frac{5}{13}$$

Now, apply the cosine rule to find c :

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Substitute the known values:

$$c^2 = 13^2 + 14^2 - 2 \times 13 \times 14 \times \left(\frac{5}{13}\right)$$

Calculate:

$$c^2 = 365 - 140 = 225$$

$$c = 15$$

Step 2: Calculate the Semi-perimeter

The semi-perimeter S is:

$$S = \frac{a+b+c}{2} = \frac{13+14+15}{2} = \frac{42}{2} = 21$$

Step 3: Find the Area using Heron's Formula

Calculate the area Δ :

$$\Delta = \sqrt{S(S-a)(S-b)(S-c)} = \sqrt{21 \times (21-13) \times (21-14) \times (21-15)}$$

$$\Delta = \sqrt{21 \times 8 \times 7 \times 6} = 84$$

Step 4: Calculate r_1

The radius r_1 of the excircle opposite A is:

$$r_1 = \frac{\Delta}{s-a} = \frac{84}{21-13} = \frac{84}{8} = 10.5$$

Multiplying by 2:

$$2r_1 = 2 \times 10.5 = 21$$

Conclusion

From the calculations, we find that $2r_1 = S$.

Question40

In $\triangle ABC$, if $(r_2 - r_1)(r_3 - r_1) = 2r_2r_3$, then $2(r + R) =$

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Options:

A. $a + b$

B. $c + a$

C. $2\sqrt{2}R \cos\left(\frac{C-A}{2}\right)$

D. $2\sqrt{2}R \cos\left(\frac{B-C}{2}\right)$

Answer: D

Solution:

Given, in a $\triangle ABC$

$$(r_2 - r_1)(r_3 - r_1) = 2r_2r_3$$

$$\Rightarrow (r_1r_2 + r_2r_3 + r_1r_3) - r_1^2 = 0$$

$$\Rightarrow \frac{r_1r_2r_3}{r} - r_1^2 = 0$$

$$\Rightarrow r_1 \left(\frac{r_2r_3}{r} - r_1 \right) = 0$$

$$\Rightarrow m_1 = r_2r_3 \quad (\because r_1 \neq 0)$$

$$\Rightarrow \frac{\Delta}{s} \cdot \frac{\Delta}{s-a} = \frac{\Delta}{s-b} \cdot \frac{\Delta}{s-c}$$

$$\Rightarrow \frac{(s-b)(s-c)}{s(s-a)} = 1$$

$$\Rightarrow \tan^2 \frac{A}{2} = \tan^2 45^\circ$$

$$\Rightarrow A = 90^\circ$$

$$\therefore \frac{B+C}{2} = 45^\circ$$

$$\text{Now, } 2(r + R) = 2 \left(4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} + R \right)$$

$$= 2R \left(4 \left(\frac{1}{\sqrt{2}} \right) \sin \frac{B}{2} \sin \frac{C}{2} + 1 \right)$$

$$= 2\sqrt{2}R \left(2 \sin \frac{B}{2} \sin \frac{C}{2} + \frac{1}{\sqrt{2}} \right)$$

$$= 2\sqrt{2}R \left(2 \sin \frac{B}{2} \sin \frac{C}{2} + \cos \left(\frac{B+C}{2} \right) \right)$$

$$= 2\sqrt{2}R \left(2 \sin \frac{B}{2} \sin \frac{C}{2} + \cos \frac{B}{2} \cos \frac{C}{2} - \sin \frac{B}{2} \sin \frac{C}{2} \right)$$

$$= 2\sqrt{2}R \left(\cos \frac{B}{2} \cos \frac{C}{2} + \sin \frac{B}{2} \sin \frac{C}{2} \right) = 2\sqrt{2}R \cos \left(\frac{B-C}{2} \right)$$

Question41

In a Δ if the angles are in the ratio 3 : 2 : 1, then the ratio of its sides is

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Options:

A. 1 : 2 : 3

B. 2 : $\sqrt{3}$: 1

C. 3 : $\sqrt{2}$: 1

D. 1 : $\sqrt{3}$: 3

Answer: B

Solution:

Given, angle ratio is 3 : 2 : 1 We need to find sides Ratio.

Let $A = 3\theta$, $B = 2\theta$ and $C = \theta$

$\therefore A + B + C = \pi$ [angle sum property]

$$3\theta + 2\theta + \theta = \pi$$

$$6\theta = \pi$$

$$\theta = \frac{\pi}{6}$$

$$A = 3 \times \frac{\pi}{6} = \frac{\pi}{2}$$

$$B = 2 \times \frac{\pi}{6} = \frac{\pi}{3}$$

$$C = \frac{\pi}{6}$$

$$a : b : c = \sin \frac{\pi}{2} : \sin \frac{\pi}{3} : \sin \frac{\pi}{6}$$

$$\Rightarrow a : b : c = 1 : \frac{\sqrt{3}}{2} : \frac{1}{2}$$

$$a : b : c = 2 : \sqrt{3} : 1$$

Question42

In a $\triangle ABC$, if $BC = 5$, $CA = 6$ and $AB = 7$, then the length of the median drawn from B onto AC is

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Options:

A. 5

B. $7\sqrt{5}$

C. $7\sqrt{2}$

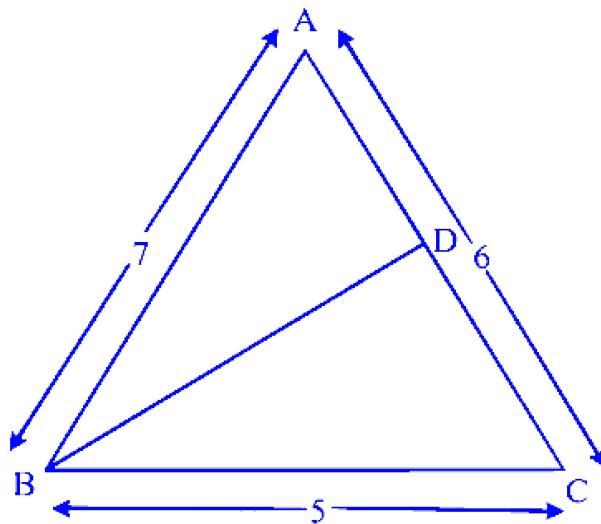
D. $2\sqrt{7}$

Answer: D

Solution:

In $\triangle ABC$, $BC = 5$, $CA = 6$, $AB = 7$.

Draw median from B onto $AC = 1$.



In $\triangle ABC$, point D is mid-point of AC .

$$AD = DC = \frac{1}{2}AC = 3$$

$$(AB)^2 + (BC)^2 = 2BD^2 + 2DA^2$$

[using Apollonius theorem]

$$(7)^2 + (5)^2 = 2BD^2 + 2 \times (3)^2$$

$$49 + 25 = 2BD^2 + 2 \times 9$$

$$74 = 2BD^2 + 18$$

$$2BD^2 = 74 - 18 = 56$$

$$BD^2 = 28$$

$$BD^2 = 7 \times 4 \Rightarrow BD = 2\sqrt{7}$$

Hence, length of median drawn from B onto

$$AC = 2\sqrt{7}$$

Question43

In $\triangle ABC$, if $AB : BC : CA = 6 : 4 : 5$, then $R : r$ is equal to

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Options:

A. $16 : 9$

B. $16 : 7$

C. $12 : 7$

D. $12 : 9$

Answer: B

Solution:

To find the ratio $R : r$ in $\triangle ABC$ with side ratios $AB : BC : CA = 6 : 4 : 5$, we use the following formula:

$$\frac{R}{r} = \frac{abc}{4(s-a)(s-b)(s-c)}$$

where s is the semi-perimeter, which can be calculated as:

$$s = \frac{a+b+c}{2}$$

Given the side ratios, let $a = 6x$, $b = 4x$, and $c = 5x$. Therefore, the semi-perimeter s is:

$$s = \frac{6x+4x+5x}{2} = \frac{15x}{2}$$

Substitute into the formula:

$$\frac{R}{r} = \frac{6x \cdot 4x \cdot 5x}{4\left(\frac{15x}{2} - 6x\right)\left(\frac{15x}{2} - 4x\right)\left(\frac{15x}{2} - 5x\right)}$$

Calculate each term:

$$s - a = \frac{15x}{2} - 6x = \frac{15x-12x}{2} = \frac{3x}{2}$$

$$s - b = \frac{15x}{2} - 4x = \frac{15x-8x}{2} = \frac{7x}{2}$$

$$s - c = \frac{15x}{2} - 5x = \frac{15x-10x}{2} = \frac{5x}{2}$$

Substitute back in:

$$\frac{R}{r} = \frac{6x \cdot 4x \cdot 5x}{4 \cdot \left(\frac{3x}{2}\right) \cdot \left(\frac{7x}{2}\right) \cdot \left(\frac{5x}{2}\right)}$$

Simplify:

$$\frac{R}{r} = \frac{120x^3}{4 \cdot \frac{3x}{2} \cdot \frac{7x}{2} \cdot \frac{5x}{2}} = \frac{120x^3}{4 \cdot \frac{105x^3}{8}}$$

$$= \frac{120x^3 \cdot 8}{4 \cdot 105x^3} = \frac{960}{420} = \frac{16}{7}$$

Thus, $R : r$ is $\frac{16}{7}$.

Question44

If (α, β) is the orthocentre of the triangle with the vertices $(2, 2), (5, 1), (4, 4)$, then $\alpha + \beta =$

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Options:

A. 6

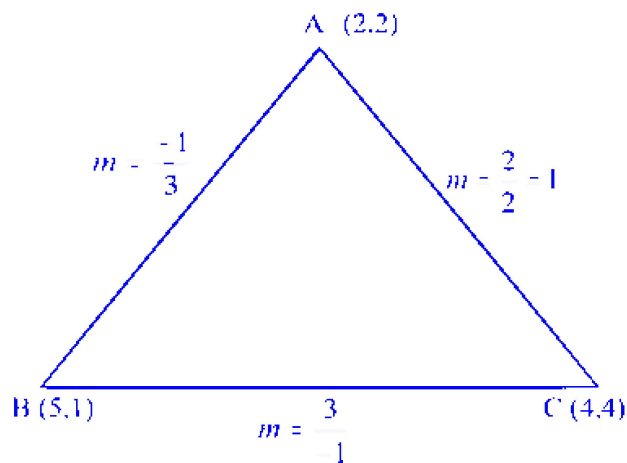
B. 5

C. $\frac{5}{2}$

D. $\frac{7}{2}$

Answer: A

Solution:



Altitude of BC,

$$y - 2 = \frac{1}{3}(x - 2)$$

$$3y - 6 = x - 2$$

$$x - 3y + 4 = 0$$

Altitude of AC, $y + 1 = -1(x - 5)$

$$y - 1 = -x + 5$$

$$x + y - 6 = 0$$

From Eqs. (i) and (ii), we get

$$-4y + 10 = 0$$

$$y = +\frac{5}{2}$$

$$\Rightarrow \beta = \frac{5}{2} \Rightarrow x + \frac{5}{2} = 6$$

$$\Rightarrow 2x = 12 - 5$$

$$x = \frac{7}{2}$$

$$\alpha = \frac{7}{2}$$

$$\Rightarrow \alpha + \beta = \frac{7}{2} + \frac{5}{2} = \frac{12}{2} = 6$$

$$\therefore \alpha + \beta = 6$$

Question45

In $\triangle ABC$, if $4r_1 = 5r_2 = 6r_3$, then $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} =$

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Options:

A. $\frac{19}{22}$

B. $\frac{25}{33}$

C. $\frac{74}{99}$

D. $\frac{28}{33}$

Answer: B

Solution:

$$\therefore \sin^2 \frac{A}{2} = \frac{(s-b)(s-c)}{bc},$$

$$\sin^2 \frac{B}{2} = \frac{(s-c)(s-a)}{ac}$$

$$\text{and } \sin^2 \frac{C}{2} = \frac{(s-a)(s-b)}{ab}$$

$$\Rightarrow \sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2}$$

$$= \frac{(s-b)(s-c)}{bc} + \frac{(s-c)(s-a)}{ac} + \frac{(s-a)(s-b)}{ab} \quad \dots \text{ (i)}$$

$$\text{Given, } 4r_1 = 5r_2 = 6r_3$$

$$\Rightarrow r_1 : r_2 = 5 : 4 \text{ and } r_2 : r_3 = 6 : 5$$

$$\Rightarrow r_1 : r_2 : r_3 = 30 : 24 : 20$$

$$\Rightarrow \frac{s}{s-a} : \frac{s}{s-b} : \frac{s}{s-c} = 30 : 24 : 20$$

$$\Rightarrow s-a : s-b : s-c = \frac{1}{30} : \frac{1}{24} : \frac{1}{20}$$

$$\therefore s-b+s-c=2s-b-c=a$$

$$= \frac{1}{24} + \frac{1}{20} = \frac{11}{120}$$

$$\text{Similarly, } b = \frac{1}{30} + \frac{1}{20} = \frac{50}{600} = \frac{10}{120}$$

$$\text{and } C = \frac{1}{30} + \frac{1}{24} = \frac{54}{720} = \frac{9}{120}$$

$$\therefore a : b : c = 11 : 10 : 9$$

$$s = \frac{a+b+c}{2} = \frac{11+10+9}{2} = 15$$

From Eq. (i), we get

$$\begin{aligned} \frac{\sin^2 A}{2} + \frac{\sin^2 B}{2} + \frac{\sin^2 C}{2} \\ = \frac{5 \times 6}{90} + \frac{6 \times 4}{99} + \frac{4 \times 5}{110} \\ \Rightarrow \frac{30}{90} + \frac{24}{99} + \frac{20}{110} = \frac{25}{33} \end{aligned}$$

Question46

$$\text{In } \triangle ABC, r_1 \cot \frac{A}{2} + r_2 \cot \frac{B}{2} + m_3 \cot \frac{C}{2} =$$

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Options:

A. 3Δ

B. $3S$

C. $\frac{s}{\Delta}$

D. Δ

Answer: A

Solution:

We know that

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c}$$

$$\cot \frac{A}{2} = \sqrt{\frac{s(s-a)}{(s-b)(s-c)}},$$

$$\cot \frac{B}{2} = \sqrt{\frac{s(s-b)}{(s-a)(s-c)}}$$

$$\cot \frac{C}{2} = \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}, r = \frac{\Delta}{s}$$

$$\text{Now, } \pi_1 \cot \frac{A}{2} + \pi_2 \cot \frac{B}{2} + \pi_3 \cot \frac{C}{2}$$

$$= \frac{\Delta}{s} \cdot \frac{\Delta}{s-a} \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \frac{\Delta}{s} \cdot \frac{\Delta}{s-b} \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} + \frac{\Delta}{s} \cdot \frac{\Delta}{s-c} \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

$$= \Delta^2 \sqrt{\frac{1}{s(s-a)(s-b)(s-c)}} + \Delta^2 \sqrt{\frac{1}{(s-a)(s-b)(s-c)s}} + \Delta^2 \sqrt{\frac{1}{(s-a)(s-b)(s-c)s}}$$

$$= 3\Delta^2 \cdot \frac{1}{\sqrt{s(s-a)(s-b)(s-c)}} = 3\Delta$$

$$rr_1 \cot \frac{A}{2} + rr_2 \cot \frac{B}{2} + rr_3 \cot \frac{C}{2} = 3\Delta$$

Question47

In $\triangle ABC$, $bc - r_2 r_3 =$

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Options:

A.

$$rr_1$$

B.

$$rr_2$$

C. r_1

D. ar_1

Answer: A

Solution:

$$\begin{aligned}
\triangle ABC, bc - r_2 r_3 &= bc - \frac{\Delta}{s-b} \cdot \frac{\Delta}{s-c} \\
&= bc - \frac{\Delta^2}{(s-b)(s-c)} \\
&= bc - \frac{s(s-a)(s-b)(s-c)}{(s-b)(s-c)} \\
&= bc - s(s-a) = (bc - s(s-a)) \\
&= bc \left[1 - \frac{s(s-a)}{bc} \right] \\
&= bc \left(1 - \cos^2 \frac{A}{2} \right) = bc \cdot \sin^2 \frac{A}{2} \\
&= bc \cdot \frac{(s-b)(s-c)}{bc} \\
&= \frac{(s-b)(s-c)(s-a) \cdot s}{s(s-a)} \\
&= \frac{\Delta^2}{s(s-a)} = \frac{\Delta}{s} \cdot \frac{\Delta}{s-a} \\
&= r \cdot r_1
\end{aligned}$$

Question48

If $O(0, 0, 0)$, $A(3, 0, 0)$ and $B(0, 4, 0)$ form a triangle, then the incentre of $\triangle OAB$ is

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Options:

- A. $(0, 1, 0)$
- B. $(0, 1, 1)$
- C. $(1, 0, 1)$
- D. $(1, 1, 0)$

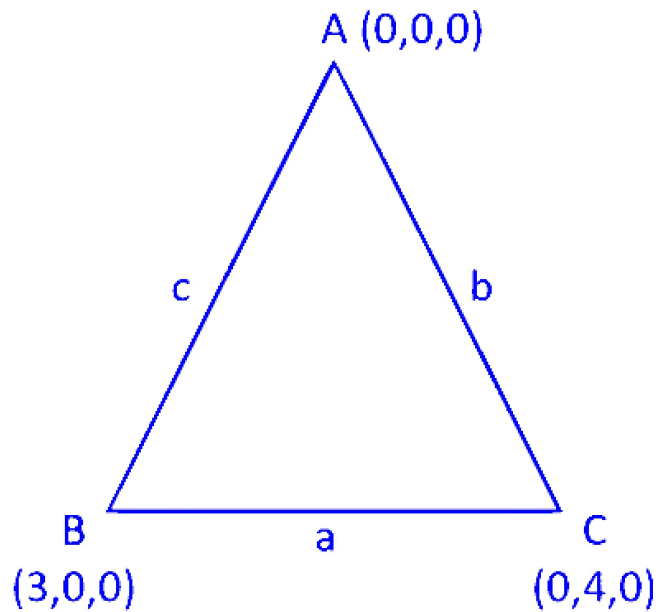
Answer: D

Solution:

The incentre of triangle formed by the points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) is

$$\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}, \frac{az_1 + bz_2 + cz_3}{a + b + c} \right)$$

where a, b, c are the sides of the triangle. The vertices are $A(0, 0, 0)$, $B(3, 0, 0)$ and $C(0, 4, 0)$



Here, the sides are

$$AB = 3, AC = 4$$

$$\text{and } BC = 5$$

The incentre of the triangle is

$$\left(\frac{5(0) + 4(3) + 3(0)}{12}, \frac{5(0) + 4(0) + 3(4)}{12}, \frac{5(0) + 4(0) + 3(0)}{12} \right) \\ = \left(\frac{12}{12}, \frac{12}{12}, 0 \right) = (1, 1, 0)$$

Question49

In $\triangle ABC$, if $r_1 = 4$, $r_2 = 8$ and $r_3 = 24$, then $a =$

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Options:

A. 0

B. $\frac{16}{\sqrt{5}}$

C. $16\sqrt{5}$

D. $\sqrt{5}$

Answer: B

Solution:

We have, $r_1 = 4, r_2 = 8$ and $r_3 = 24$

$$\begin{aligned}\frac{1}{r} &= \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{4} + \frac{1}{8} + \frac{1}{24} \\ &= \frac{6+3+1}{24} = \frac{10}{24} = \frac{5}{12} \Rightarrow r = \frac{12}{5}\end{aligned}$$

$$\Delta^2 = r_1 \cdot r_2 \cdot r_3 \cdot r = 4 \times 8 \times 24 \times \frac{12}{5}$$

$$\text{So, } \Delta = \frac{96}{\sqrt{5}} \Rightarrow r_1 = \frac{\Delta}{s-a}$$

$$\Rightarrow s - a = \frac{\Delta}{r_1} = \frac{96}{\sqrt{5} \times 4} = \frac{24}{\sqrt{5}}$$

$$\text{Similarly, } s - b = \frac{\Delta}{r_2} = \frac{96}{\sqrt{5} \times 8} = \frac{12}{\sqrt{5}} \dots$$

$$s - c = \frac{\Delta}{r_3} = \frac{96}{\sqrt{5} \times 24} = \frac{4}{\sqrt{5}} \dots$$

On adding Eqs. (i), (ii) and (iii), we get

$$s - a + s - b + s - c = \frac{24}{\sqrt{5}} + \frac{12}{\sqrt{5}} + \frac{4}{\sqrt{5}}$$

$$3s - (a + b + c) = \frac{40}{\sqrt{5}}$$

$$3s - 2s = \frac{40}{\sqrt{5}} \Rightarrow s = \frac{40}{\sqrt{5}}$$

$$\text{Then, } a = s - \frac{24}{\sqrt{5}} = \frac{40}{\sqrt{5}} - \frac{24}{\sqrt{5}} = \frac{16}{\sqrt{5}}$$

Question 50

Match the items of List I with those of List II (here, Δ denotes the area of $\triangle ABC$)

List I		List II	
(A)	$\sum \cot A$	(i)	$(a + b + c)^2 \frac{1}{4\Delta}$
(B)	$\sum \cot \frac{A}{2}$	(ii)	$(a^2 + b^2 + c^2) \frac{1}{4\Delta}$
(C)	If $\tan A : \tan B : \tan C = 1 : 2 : 3$, then $\sin A : \sin B : \sin C =$	(iii)	$8 : 6 : 5$

List I		List II	
(D)	If $\cot \frac{A}{2} : \cot \frac{B}{2} : \cot \frac{C}{2} = 3 : 7 : 9$ then $a : b : c =$	(iv)	$12 : 5 : 13$
		(v)	$\sqrt{5} : 2\sqrt{2} : 3$
		(vi)	4Δ

Then, the correct match is

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Options:

A. A-VI, B-I, C-II, D-III

B. A-II, B-I, C-V, D-III

C. A-II, B-VI, C-V, D-I

D. A-VI, B-II, C-I, D-IV

Answer: B

Solution:

Let a, b, c be the sides of $\triangle ABC$, tha

$$\cot A = \frac{b^2 + c^2 - a^2}{4\Delta}$$

$$\cot B = \frac{a^2 + c^2 - b^2}{4\Delta}, \cot C = \frac{a^2 + b^2 - c^2}{4\Delta}$$

$$\text{So, } \Sigma \cot A = \cot A + \cot B + \cot C = \frac{1}{45}$$

$$\begin{aligned} & [b^2 + c^2 - a^2 + a^2 + c^2 - b^2 + a^2 + b^2 - \\ & = \frac{a^2 + b^2 + c^2}{4\Delta} \end{aligned}$$

(B) In $\triangle ABC$,

$$\cot \frac{A}{2} = \frac{s(s-a)}{\sqrt{s(s-a)(s-b)(s-c)}} = \frac{s(s-c)}{\Delta}$$

$$\cot \frac{B}{2} = \frac{s-b}{\Delta}$$

$$\cot \frac{C}{2} = \frac{s-c}{\Delta} \text{ where, } s = \frac{a+b+c}{2}$$

$$\Sigma \cot \frac{A}{2} = \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}$$

$$\begin{aligned} \text{Thus, } &= \frac{s(s-a)}{\Delta} + \frac{s(s-b)}{\Delta} + \frac{s(s-c)}{\Delta} \\ &= \frac{s}{\Delta} [s-a + s-b + s-c] \end{aligned}$$

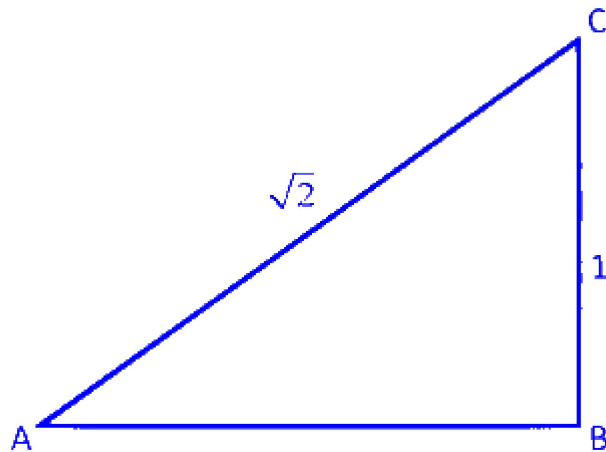
$$= \frac{s}{\Delta} [3s - (a+b+c)]$$

$$= \frac{s}{\Delta} (3s - 2s) = \frac{s^2}{\Delta} = \frac{(a+b+c)^2}{4\Delta}$$

(C) We have, $\tan A : \tan B : \tan C = 1 : 2 : 3$

So, $\tan A = k, \tan B = 2K, \tan C = 3K$

Now, $\tan C = \tan[\pi - (A + B)] = 3K$



$$\Rightarrow -\tan(A+B) = 3K$$

$$\Rightarrow \tan(A+B) = -3K$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = -3K$$

$$\Rightarrow \frac{K+2K}{1-2K^2} = -3K$$

$$\Rightarrow 1 - 2K^2 = -1 \Rightarrow 2K^2 = 2$$

$$\Rightarrow K^2 = 1 \Rightarrow K = 1$$

$$\sin A = \frac{1}{\sqrt{2}} \Rightarrow \sin B = \frac{2}{\sqrt{5}}$$

$$\sin C = \frac{3}{\sqrt{10}}$$

$$\sin A : \sin B : \sin C$$

$$= \frac{1}{\sqrt{2}} : \frac{2}{\sqrt{5}} : \frac{3}{\sqrt{10}} = \sqrt{5} : 2\sqrt{2} : 3$$

(D) We have,

$$\cot\left(\frac{A}{2}\right) : \cot\left(\frac{B}{2}\right) : \cot\left(\frac{C}{2}\right) = 3 : 5 : 7$$

$$\Rightarrow \frac{\cot A/2}{3} = \frac{\cot B/2}{7} = \frac{\cot C/2}{9} = k$$

$$\Rightarrow \frac{s(s-a)}{3\Delta} = \frac{s(s-b)}{7\Delta} = \frac{s(s-c)}{9\Delta}$$

$$\Rightarrow \frac{s-a}{3} = \frac{s-b}{7} = \frac{s-c}{9}$$

$$\Rightarrow \frac{s-b+s-c}{9+7} = \frac{s-a+s-c}{3+9}$$

$$\Rightarrow = \frac{s-a+s-b}{3+7}$$

$$\Rightarrow \frac{2s-b}{16} = \frac{a}{16} = \frac{b}{12} = \frac{c}{10} = K$$

$$\Rightarrow a : b : c = 16 : 12 : 10 = 8 : 6 : 5$$

Question51

In a $\triangle ABC$, if $r_1 = 2r_2 = 3r_3$, then $\sin A : \sin B : \sin C =$

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Options:

A. $5 : 4 : 2$

B. $3 : 4 : 2$

C. $6 : 3 : 2$

D. $5 : 4 : 3$

Answer: D

Solution:

$$\begin{aligned}
r_1 &= 2r_2 = 3r_3 \\
\Rightarrow \frac{\Delta}{s-a} &= \frac{2\Delta}{s-b} = \frac{3\Delta}{s-c} \\
\Rightarrow \frac{1}{s-a} &= \frac{2}{s-b} = \frac{3}{s-c} = k \text{ (let)} \\
\Rightarrow s-a &= \frac{1}{k}, s-b = \frac{2}{k} \text{ and } s-c = \frac{3}{k}, s = \frac{6}{k} \\
\Rightarrow a &= s - (s-a) = \frac{5}{k} \\
\text{Similarly, } b &= \frac{4}{k} \text{ and } c = \frac{3}{k} \\
\text{We know that } \frac{\sin A}{a} &= \frac{\sin B}{b} = \frac{\sin C}{c} \\
\Rightarrow \frac{\sin A}{5} &= \frac{\sin B}{4} = \frac{\sin C}{3} \\
\Rightarrow \sin A : \sin B : \sin C &= 5 : 4 : 3
\end{aligned}$$

Question 52

In $\triangle ABC$, if $B = 90^\circ$, then $2(r + R) =$

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Options:

A. $a + b$

B. $b + c$

C. $a + c$

D. 0

Answer: C

Solution:

The circumradius R for a right-angled triangle is given by:

$$\frac{b}{\sin B} = 2R$$

Since $\sin 90^\circ = 1$, we have:

$$b = 2R$$

Therefore, $R = \frac{b}{2}$.

The inradius r is given by:

$$r = (s - b) \tan \frac{B}{2} = (s - b)$$

Since $\angle B = 90^\circ$, $\frac{B}{2} = 45^\circ$ and $\tan 45^\circ = 1$. Hence:

$$r = s - b$$

Now, we compute $2(r + R)$:

$$2(r + R) = 2\left(s - b + \frac{b}{2}\right) = 2s - b$$

Further simplifying, considering the semi-perimeter $s = \frac{a+b+c}{2}$:

$$2s = a + b + c$$

Therefore, substituting back:

$$2s - b = a + b + c - b = a + c$$

So, the value is $a + c$.

Question53

In a $\triangle ABC$, if $(a - b)(s - c) = (b - c)(s - a)$, then $r_1 + r_3 =$

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Options:

A. $r_2 - r_3$

B. $2r_2$

C. $3r_2$

D. $3(r_1 + r_2)$

Answer: B

Solution:

In a triangle $\triangle ABC$, we have the relation:

$$(a - b)(s - c) = (b - c)(s - a)$$

Rearranging the terms, we obtain:

$$(s - b)(s - c) + (s - b)(s - a) = 2(s - a)(s - c)$$

Next, divide each term by $(s - a)(s - b)(s - c)$ and multiply by the area of the triangle Δ :

$$\frac{\Delta}{s-a} + \frac{\Delta}{s-c} = \frac{2\Delta}{s-b}$$

This simplifies to:

$$r_1 + r_3 = 2r_2$$

Question54

In $\triangle ABC$, $\cos A + \cos B + \cos C =$

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Options:

A. $\frac{1+\sqrt{2}}{R}$

B. $\frac{1}{R}$

C. $\frac{1+R}{R}$

D.

$1 + \frac{r}{R}$

Answer: D

Solution:

We have,

ABC is a triangle

$$\cos A + \cos B + \cos C$$

$$\begin{aligned}
&= 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) + 1 - 2 \sin^2 \frac{C}{2} \\
&= 1 + 2 \sin \frac{C}{2} \cos \left(\frac{A-B}{2} \right) - 2 \sin^2 \frac{C}{2} \quad [\because A+B+C=\pi] \\
&= 1 + 2 \sin \frac{C}{2} \left[\cos \left(\frac{A-B}{2} \right) - \sin \frac{C}{2} \right] \\
&= 1 + 2 \sin \frac{C}{2} \left[\cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right] \\
&= 1 + 2 \sin \frac{C}{2} \left[2 \sin \frac{A}{2} \sin \frac{B}{2} \right] \\
&= 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\
&= 1 + 4 \left[\sqrt{\frac{(s-b)(s-c)}{bc}} \sqrt{\frac{(s-a)(s-c)}{ac}} \right] \\
&= \frac{1 + 4(s-a)(s-b)(s-c)}{abc} \\
&= \frac{1 + 4s(s-a)(s-b)(s-c)}{sabc} [\because \text{multiply and divide by } s] \\
&= 1 + 4 \frac{\Delta^2}{sabc}, \text{ where } \Delta \text{ is area of triangle} \\
&= 1 + \left(\frac{\Delta}{s} \right) \left(\frac{4\Delta}{abc} \right) \quad \left[\because r = \frac{\Delta}{s} \text{ and } R = \frac{abc}{4\Delta} \right] \\
&= 1 + \frac{r}{R}
\end{aligned}$$

Question55

In a $\triangle ABC$, if $a = 26$, $b = 30$, $\cos c = \frac{63}{65}$, then $c =$

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Options:

- A. 2
- B. 4
- C. 6
- D. 8

Answer: D

Solution:

In triangle $\triangle ABC$, given $a = 26$, $b = 30$, and $\cos C = \frac{63}{65}$, we need to find the length of side c .

We know the cosine rule for any triangle is given by:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Substituting the known values:

$$\frac{a^2 + b^2 - c^2}{2 \times a \times b} = \frac{63}{65}$$

This implied expression becomes:

$$\frac{26^2 + 30^2 - c^2}{2 \times 26 \times 30} = \frac{63}{65}$$

Now, solve this step-by-step:

Calculate 26^2 and 30^2 :

$$26^2 = 676, \quad 30^2 = 900$$

Substitute these into the equation:

$$\frac{676 + 900 - c^2}{1560} = \frac{63}{65}$$

Simplify the numerator:

$$676 + 900 = 1576$$

Substitute back:

$$\frac{1576 - c^2}{1560} = \frac{63}{65}$$

Cross-multiply to solve for c^2 :

$$65(1576 - c^2) = 63 \times 1560$$

Compute 63×1560 :

$$63 \times 1560 = 98280$$

Now, solve for c^2 :

$$65 \times (1576 - c^2) = 98280$$

Simplify the expressions:

$$102440 - 65c^2 = 98280$$

Rearrange to find c^2 :

$$102440 - 98280 = 65c^2$$

Solve:

$$4160 = 65c^2$$

Divide by 65 to isolate c^2 :

$$c^2 = \frac{4160}{65} = 64$$

Take the square root:

$$c = \sqrt{64} = 8$$

Thus, the value of c is 8.

Question 56

If H is orthocentre of $\triangle ABC$ and $AH = x$; $BH = y$; $CH = z$, then $\frac{abc}{xyz} =$

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Options:

A. 1

B. $\frac{a+b+c}{x+y+z}$

C. $\frac{a}{x} + \frac{b}{y} + \frac{c}{z}$

D. $\frac{ab+bc+ca}{xy+yz+zx}$

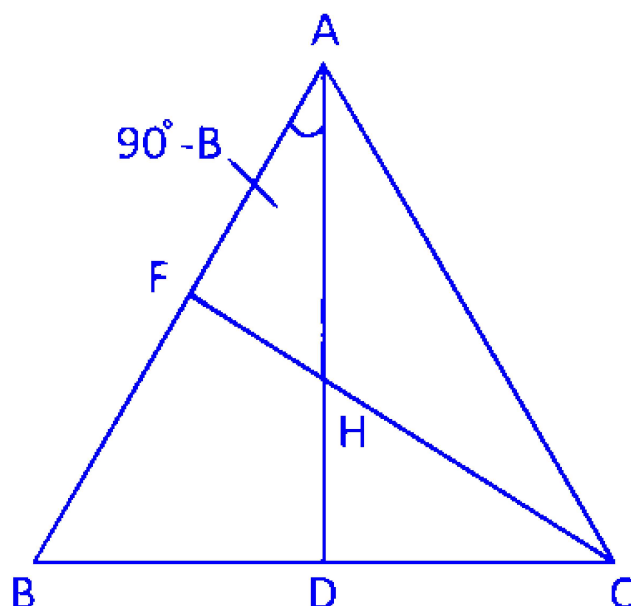
Answer: C

Solution:

In $\triangle AFC$

$$\cos A = \frac{AF}{AC} = \frac{AF}{b}$$

$$\Rightarrow AF = b \cos A$$



In $\triangle AFH$

$$\cos(90^\circ - 2B) = \frac{AF}{AH} \Rightarrow \sin B = \frac{b \cos A}{x}$$

$$\Rightarrow x = \left(\frac{b}{\sin B} \right) \cos A$$

Using sine rule, we get

$$x = \left(\frac{a}{\sin A} \right) \cos A$$

$$\Rightarrow \frac{a}{x} = \tan A$$

Similarly, $\frac{b}{y} = \tan B$ and $\frac{c}{z} = \tan C$

$$\begin{aligned} \therefore \frac{akc}{xyz} &= \tan A \tan B \tan C \\ &= \tan A + \tan B + \tan C \quad [\because A + B + C = \pi] \\ &= \frac{a}{x} + \frac{b}{y} + \frac{c}{z} \end{aligned}$$

Question 57

In any $\triangle ABC$, $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} =$

Options:

A. $a^2 + b^2 + c^2$

B. $\frac{a^2+b^2+c^2}{2abc}$

C. $\frac{2abc}{a^2+b^2+c^2}$

D. $a + b + c$

Answer: B

Solution:

Given, in a $\triangle ABC$,

$$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$$

We know that,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\begin{aligned} \text{Since, } & \frac{b^2 + c^2 - a^2}{2abc} + \frac{a^2 + c^2 - b^2}{2abc} + \frac{a^2 + b^2 - c^2}{2ab} \\ &= \frac{b^2 + c^2 - a^2 + a^2 + c^2 - b^2 + a^2 + b^2 - c^2}{2abc} \\ &= \frac{a^2 + b^2 + c^2}{2abc}. \end{aligned}$$

Question58

In a $\triangle ABC$, if $r_1 = 36$, $r_2 = 18$ and $r_3 = 12$, then $s =$

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Options:

A. 6

B. 8

C. 16

D. 36

Answer: D

Solution:

Given,

$$r_1 = 36, r_2 = 18 \text{ and } r_3 = 12$$

We know that, in a $\triangle ABC$

$$\begin{aligned}\frac{1}{r} &= \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \\ \Rightarrow \frac{1}{r} &= \frac{1}{36} + \frac{1}{18} + \frac{1}{12} \Rightarrow \frac{1}{r} = \frac{2+4+6}{72} \\ r &= 6\end{aligned}$$

Now, we know that,

$$\Delta^2 = r \cdot r_1 \cdot r_2 \cdot r_3 = 6 \times 36 \times 18 \times 12$$

$$\Delta^2 = 6^2 \times 6^2 \times 6^2$$

$$\Delta = 6 \times 6 \times 6 = 216$$

Again we know that,

$$\begin{aligned}r &= \frac{\Delta}{S} = \frac{216}{S} \\ \Rightarrow S &= \frac{216}{6} = 36\end{aligned}$$

Question59

In a $\triangle ABC$, $a = 6$, $b = 5$ and $c = 4$, then $\cos 2A =$

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Options:

A. $-\frac{31}{32}$

B. $-\frac{15}{16}$

C. $\frac{31}{32}$

D. $\frac{15}{16}$

Answer: A

Solution:

In a $\triangle ABC$, $a = 6$ units, $b = 5$ units and $c = 4$ units

As we know that,

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ \Rightarrow \cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 4^2 - 6^2}{2 \times 5 \times 4} = \frac{1}{8}\end{aligned}$$

As we know that,

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 = 2 \times \left(\frac{1}{8}\right)^2 - 1 \\ &= \frac{1}{32} - 1 = \frac{-31}{32}\end{aligned}$$

Question60

In a $\triangle ABC$, $\left(\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}\right)^2 \leq$

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Options:

A. $\frac{1}{27}$

B. $\frac{1}{18}$

C. $\frac{1}{9}$

D. $\frac{1}{3}$

Answer: A

Solution:

Within the context of triangle $\triangle ABC$, the following relationship holds:

We know that:

$$\left(\tan \frac{A}{2} \cdot \tan \frac{B}{2} \cdot \tan \frac{C}{2}\right) \leq \frac{1}{3\sqrt{3}}$$

Thus, squaring both sides of the inequality, we get:

$$\begin{aligned}\left(\tan \frac{A}{2} \cdot \tan \frac{B}{2} \cdot \tan \frac{C}{2}\right)^2 &\leq \left(\frac{1}{3\sqrt{3}}\right)^2 \\ &= \frac{1}{27}\end{aligned}$$

Question61

In a $\triangle ABC$, $2(bc \cos A + ac \cos B + ab \cos C) =$

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Options:

A. $a + b + c$

B. $2(a + b + c)$

C. $a^2 + b^2 + c^2$

D. $2(a^2 + b^2 + c^2)$

Answer: C

Solution:

Given that ABC be triangle.

Using cosine rule, $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca}, \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Now, $2(bc \cos A + ca \cos B + ab \cos C)$

$$\begin{aligned}
&= 2 \left(bc \frac{b^2+c^2-a^2}{2bc} + ca \frac{c^2+a^2-b^2}{2ca} + ab \frac{a^2+b^2-c^2}{2ab} \right) \\
&= 2 \left(\frac{b^2+c^2-a^2}{2} + \frac{c^2+a^2-b^2}{2} + \frac{a^2+b^2-c^2}{2} \right) \\
&= 2 \left(\frac{b^2+c^2+a^2}{2} \right) = a^2 + b^2 + c^2
\end{aligned}$$

Question62

In a $\triangle ABC$, $\frac{a}{b} = 2 + \sqrt{3}$ and $\angle C = 60^\circ$. Then, the measure of $\angle A$ is

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Options:

A. 95°

B. 65°

C. 105°

D. 115°

Answer: C

Solution:

Given that ABC is a triangle

So, $A + B + C = 180^\circ$

Given, $C = 60^\circ$

$\Rightarrow A + B = 120^\circ \quad \dots (i)$

We have by sine formula

$$\begin{aligned}
\frac{\sin A}{a} &= \frac{\sin B}{b} \\
\Rightarrow \frac{\sin A}{\sin B} &= \frac{a}{b} = 2 + \sqrt{3} \text{ (given)}
\end{aligned}$$

Applying componendo and dividendo, we get

$$\begin{aligned}
\frac{\sin A + \sin B}{\sin A - \sin B} &= \frac{2 - \sqrt{3} + 1}{2 + \sqrt{3} - 1} = \frac{3 + \sqrt{3}}{\sqrt{3} + 1} \\
\Rightarrow \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)} &= \frac{(3 + \sqrt{3})(\sqrt{3} - 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} \\
\Rightarrow \frac{2 \left(\frac{\sqrt{3}}{2} \right)}{2 \left(\frac{1}{2} \right)} \cot \left(\frac{A-B}{2} \right) &= \sqrt{3} \\
\Rightarrow \sqrt{3} \cot \left(\frac{A-B}{2} \right) = \sqrt{3} \Rightarrow \left(\frac{A-B}{2} \right) &= 45^\circ \\
\Rightarrow A - B = 90 \quad \dots (ii)
\end{aligned}$$

Solving Eqs. (i) and (ii), we get $\angle A = 105^\circ$

Question63

If $a = 2, b = 3, c = 4$ in a $\triangle ABC$, then $\cos C =$

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Options:

- A. $\frac{1}{4}$
- B. $\frac{-1}{4}$
- C. $\frac{1}{2}$
- D. $\frac{-1}{2}$

Answer: B

Solution:

Given, $a = 2, b = 3, c = 4$

Thus,

$$\begin{aligned}
\cos C &= \frac{a^2 + b^2 - c^2}{2ab} \\
&= \frac{4 + 9 - 16}{2 \times 2 \times 3} = \frac{-3}{12} = \frac{-1}{4}
\end{aligned}$$

Question64

In a $\triangle ABC$ $(b + c) \cos A + (c + a) \cos B + (a + b) \cos C =$

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Options:

A. $2abc$

B. abc

C. $a + b + c$

D. $(a + b + c)/2abc$

Answer: C

Solution:

$$\begin{aligned}(b + c) \cos A + (c + a) \cos B + (a + b) \cos C \\&= b \cos A + c \cos A + c \cos B + a \cos B + a \cos C + b \cos C \\&= (b \cos C + c \cos B) + (c \cos A + a \cos C) \\&\quad + (a \cos B + b \cos A) \quad \dots \text{ (i)}\end{aligned}$$

Using projection formula,

$$a = (b \cos C + c \cos B)$$

$$b = (c \cos A + a \cos C)$$

$$c = (a \cos B + b \cos A)$$

Substituting above values in Eq. (i), we get

$$a + b + c$$

$$\text{Thus, } (b + c) \cos A + (c + a) \cos B + (a + b) \cos C = a + b + c$$

Question65

Suppose $\triangle ABC$ is an isosceles triangle with $\angle C = 90^\circ$, $A = (2, 3)$ and $B = (4, 5)$. Then, the centroid of the triangle is

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Options:

A. $\left(\frac{13}{8}, \frac{8}{3}\right)$

B. $\left(\frac{11}{3}, \frac{10}{3}\right)$

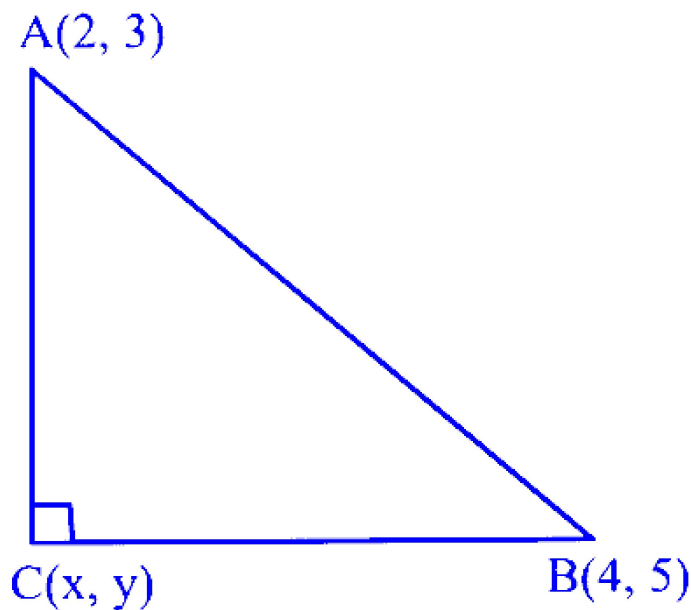
C. $\left(\frac{10}{3}, \frac{13}{3}\right)$

D. $\left(\frac{10}{3}, \frac{11}{3}\right)$

Answer: D

Solution:

Let the coordinates of C be (x, y).



$$\text{Slope of } AC \times \text{Slope of } BC = -1$$

$$\left(\frac{y-3}{x-2}\right) \times \left(\frac{y-5}{x-4}\right) = -1$$

$$\Rightarrow y^2 - 8y + 15 = -x^2 + 6x - 8$$

$$\Rightarrow x^2 + y^2 - 6x - 8y + 23 = 0 \quad \dots (i)$$

$$(AC)^2 = (BC)^2 \text{ [ABC is an isosceles right angled triangle]}$$

$$\begin{aligned}
&\Rightarrow (x-2)^2 + (y-3)^2 = (x-4)^2 + (y-5)^2 \\
&\Rightarrow x^2 + 4 - 4x + y^2 + 9 - 6y \\
&\quad = x^2 + 16 - 8x + y^2 + 25 - 10y \\
&\Rightarrow 4x + 4y = 28 \Rightarrow x + y = 7 \\
&\Rightarrow y = 7 - x \quad \dots \text{(ii)}
\end{aligned}$$

Substituting the value of y from Eq. (ii) in the Eq. (i),

$$\begin{aligned}
&x^2 + (7-x)^2 - 6x - 8(7-x) + 23 = 0 \\
&\Rightarrow x^2 + 49 + x^2 - 14x - 6x - 56 + 8x + 23 = 0 \\
&\Rightarrow 2x^2 - 12x + 16 = 0 \Rightarrow x^2 - 6x + 8 = 0 \\
&\Rightarrow x = 4, 2
\end{aligned}$$

If $x = 4$, then $y = 3$ and If $x = 2$, then $y = 5$. If $(x, y) = (4, 3)$, then

Centroid of triangle is $\left(\frac{2+4+4}{3}, \frac{3+3+5}{3}\right) = \left(\frac{10}{3}, \frac{11}{3}\right)$

If $(x, y) = (2, 5)$, then centroid of $\triangle ABC$ is

$$\left(\frac{2+2+4}{3}, \frac{5+3+5}{3}\right) = \left(\frac{8}{3}, \frac{13}{3}\right)$$

Question66

In a $\triangle ABC$, if $a \neq b$, $\frac{a \cos A - b \cos B}{a \cos B - b \cos A} + \cos C =$

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Options:

A. 0

B. 1

C. 2

D. -1

Answer: A

Solution:

$$\begin{aligned}
& \text{In } \triangle ABC, a \neq b \\
& \frac{a \cdot \cos A - b \cos B}{a \cos B - b \cos A} + \cos C \\
&= \frac{a \left[\frac{b^2+c^2-a^2}{2bc} \right] - b \left[\frac{a^2+c^2-b^2}{2ac} \right]}{a \left[\frac{a^2+c^2-b^2}{2ac} \right] - b \left[\frac{b^2+c^2-a^2}{2bc} \right]} + \cos C \\
&= \frac{a^2(b^2+c^2-a^2) - b^2(a^2+c^2-b^2)}{ab(a^2+c^2-b^2) - ab(b^2+c^2-a^2)} + \cos C \\
&= \frac{a^2b^2 + a^2c^2 - a^4 - b^2a^2 - b^2c^2 + b^4}{a^3b + abc^2 - ab^3 - ab^3 - abc^2 + a^3b} + \cos C \\
&= \frac{a^2 + b^2 - c^2}{2ab} \\
&= \frac{a^2c^2 - a^4 - b^2c^2 + b^4}{2a^3b - 2ab^3} + \frac{a^2 + b^2 - c^2}{2ab} \\
&= \frac{a^2c^2 - a^4 - b^2c^2 + b^4}{2ab(a^2 - b^2)} + \frac{(a^2 - b^2)(a^2 + b^2 - c^2)}{2ab(a^2 - b^2)} \\
&= \frac{a^2c^2 - a^4 - b^2c^2 + b^4}{2ab(a^2 - b^2)} \\
&+ \frac{a^4 + a^2b^2 - a^2c^2 - b^2a^2 - b^4 + b^2c^2}{2ab(a^2 - b^2)} \\
&= \frac{a^2c^2 - a^4 - b^2c^2 + b^4 + a^4 + a^2b^2 - a^2c^2 - b^2a^2 - b^4 + b^2c^2}{2ab(a^2 - b^2)} \\
&= \frac{0}{2ab(a^2 - b^2)} = 0
\end{aligned}$$

Question 67

If in a $\triangle ABC$, $a = 2$, $b = 3$ and $c = 4$, then $\tan(A/2) =$

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Options:

A. $\sqrt{\frac{3}{15}}$

B. $\sqrt{\frac{4}{15}}$

C. $\sqrt{\frac{2}{15}}$

D. $\sqrt{\frac{1}{15}}$

Answer: D

Solution:

In a triangle $\triangle ABC$, the given sides are $a = 2$, $b = 3$, and $c = 4$.

We start by calculating the semi-perimeter (s) of the triangle:

$$s = \frac{a+b+c}{2} = \frac{2+3+4}{2} = \frac{9}{2}$$

Next, we use the formula for $\tan\left(\frac{A}{2}\right)$:

$$\tan\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

Substitute the values:

$$\tan\left(\frac{A}{2}\right) = \sqrt{\frac{\left(\frac{9}{2}-3\right)\left(\frac{9}{2}-4\right)}{\frac{9}{2}\left(\frac{9}{2}-2\right)}} = \sqrt{\frac{\left(\frac{9}{2}-\frac{6}{2}\right)\left(\frac{9}{2}-\frac{8}{2}\right)}{\frac{9}{2}\left(\frac{9}{2}-\frac{4}{2}\right)}}$$

Simplify the fractions:

$$\tan\left(\frac{A}{2}\right) = \sqrt{\frac{(9-6)(9-8)}{9(9-4)}} = \sqrt{\frac{3 \times 1}{9 \times 5}}$$

Finally, we have:

$$\tan\left(\frac{A}{2}\right) = \sqrt{\frac{1}{15}}$$

Question68

If the angles of a $\triangle ABC$ are in the ratio 1 : 2 : 3, then the corresponding sides are in the ratio

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Options:

A. $\sqrt{3} : 2 : 1$

B. $1 : \sqrt{3} : 2$

C. $\sqrt{3} : 1 : 2$

D. $1 : 2 : \sqrt{3}$

Answer: B

Solution:

The angles of $\triangle ABC$ are in ratio $1 : 2 : 3$.

Let $A = \theta, B = 2\theta, C = 3\theta$

$$\begin{aligned}\because A + B + C &= 180^\circ \\ \Rightarrow \theta + 2\theta + 3\theta &= 180^\circ \\ \Rightarrow 6\theta &= 180^\circ \\ \Rightarrow \theta &= 30^\circ\end{aligned}$$

$\therefore A = 30^\circ, B = 60^\circ$ and $C = 90^\circ$

$$\because \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$$

$a = k \sin A, b = k \sin B$ and $c = k \sin C$

$a : b : c = k \sin 30^\circ : k \sin 60^\circ : k \sin 90^\circ$

$$\begin{aligned}&= \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 \\ &= 1 : \sqrt{3} : 2\end{aligned}$$

Question 69

In a $\triangle ABC, r_1 \cot \frac{A}{2} + r_2 \cot \frac{B}{2} + r_3 \cot \frac{C}{2} =$

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Options:

A. s

B. $2s$

C. $3s$

D. $s/2$

Answer: C

Solution:

We know, $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$ and $r_3 = \frac{\Delta}{s-c}$

Also, $\cot\left(\frac{A}{2}\right) = \frac{s(s-a)}{\Delta}$, $\cot\left(\frac{B}{2}\right) = \frac{s(s-b)}{\Delta}$

and $\cot\left(\frac{C}{2}\right) = \frac{s(s-c)}{\Delta}$

$\therefore r_1 \cot\left(\frac{A}{2}\right) + r_2 \cot\left(\frac{B}{2}\right) + r_3 \cot\left(\frac{C}{2}\right)$

$$= \frac{\Delta}{(s-a)} \cdot \frac{s(s-a)}{\Delta} + \frac{\Delta}{(s-b)} \cdot \frac{s(s-b)}{\Delta} + \frac{\Delta}{(s-c)} \cdot \frac{s(s-c)}{\Delta}$$

$$= s + s + s$$

$$= 3s$$

Question70

What is the value of $(a-b)^2 \cos^2 \frac{c}{2} + (a+b)^2 \sin^2 \frac{c}{2}$ is equal to

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Options:

A. c^2

B. $a^2 + b^2$

C. $a^2 + b^2 + c^2$

D. $a^2 - b^2 + c^2$

Answer: A

Solution:

$$\begin{aligned}
& (a-b)^2 \cos^2 \frac{C}{2} + (a+b)^2 \sin^2 \frac{C}{2} \\
&= (a^2 + b^2) \left(\cos^2 \frac{C}{2} + \sin^2 \frac{C}{2} \right) + 2ab \left(\sin^2 \frac{C}{2} - \cos^2 \frac{C}{2} \right) \\
&= a^2 + b^2 + 2ab \left(1 - 2 \cos^2 \frac{C}{2} \right) \\
&= a^2 + b^2 + 2ab(-\cos C) \\
&= a^2 + b^2 - 2ab \cos C \\
&= a^2 + b^2 - 2ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \\
&= a^2 + b^2 - a^2 - b^2 + c^2 = c^2 \\
\therefore (a-b)^2 \cos^2 \frac{C}{2} + (a+b)^2 \sin^2 \frac{C}{2} &= c^2
\end{aligned}$$

Question 71

In $\triangle ABC$, suppose the radius of the circle opposite to an angle A is denoted by r_1 , similarly $r_2 \leftrightarrow \text{angle } B$, $r_3 \leftrightarrow \text{angle } C$. If $r_1 = 2$, $r_2 = 3$ and $r_3 = 6$, then what is (a, b, c) is equal to

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Options:

- A. (3, 4, 5)
- B. (3, 5, 4)
- C. (5, 4, 3)
- D. (5, 3, 4)

Answer: A

Solution:

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b} \text{ and } r_3 = \frac{\Delta}{s-c} \dots\dots (i)$$

Also,

$$\Delta^2 = r r_1 r_2 r_3 \dots\dots (ii)$$

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \dots\dots (iii)$$

From Eq. (iii), we get

$$\frac{1}{r} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = 1$$

$$\therefore r = 1$$

From Eq. (ii), we get $\Delta^2 = (1)(2)(3)(6) = 36$

$$\Rightarrow \Delta = 6$$

$$\text{Also, } r = \frac{\Delta}{s}$$

$$\Rightarrow l = \frac{6}{s} \Rightarrow s = 6$$

$$r_1 = \frac{\Delta}{s-a}$$

$$\Rightarrow s-a = \frac{\Delta}{r_1} \Rightarrow a = s - \frac{\Delta}{r_1}$$

$$\Rightarrow a = 6 - \frac{6}{2} = 6 - 3$$

$$\Rightarrow a = 3 \Rightarrow r_2 = \frac{\Delta}{s-b}$$

$$\Rightarrow b = s - \frac{\Delta}{r_2} = 6 - \frac{6}{3} \Rightarrow b = 4$$

$$r_3 = \frac{\Delta}{s-c}$$

$$\Rightarrow c = s - \frac{\Delta}{r_3} = 6 - \frac{6}{6} = 6 - 1$$

$$\Rightarrow c = 5$$

$$\therefore (a, b, c) = (3, 4, 5)$$

Question 72

If in $\triangle ABC$, $a \tan A + b \tan B = (a+b) \cdot \tan \left(\frac{A+B}{2} \right)$, then which of the following holds?

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Options:

A. $A = B$

B. $A = 2B$

C. $A = \frac{1}{2}B$

D. $A > B$

Answer: A

Solution:

$$\begin{aligned}a \tan A + b \tan B &= (a + b) \tan \left(\frac{A + B}{2} \right) \\ \Rightarrow a \left(\tan A - \tan \left(\frac{A + B}{2} \right) \right) &= b \left(\tan \left(\frac{A + B}{2} \right) - \tan B \right) \\ \Rightarrow a \left(\frac{\sin A}{\cos A} - \frac{\sin \left(\frac{A+B}{2} \right)}{\cos \left(\frac{A+B}{2} \right)} \right) &= b \left(\frac{\sin \left(\frac{A+B}{2} \right)}{\cos \left(\frac{A+B}{2} \right)} - \frac{\sin B}{\cos B} \right) \\ \Rightarrow a \left[\frac{\sin \left(A - \frac{A+B}{2} \right)}{\cos A \cos \left(\frac{A+B}{2} \right)} \right] &= b \left[\frac{\sin \left(\frac{A+B}{2} - B \right)}{\cos B \cos \left(\frac{A+B}{2} \right)} \right] \\ a \left[\frac{\sin \frac{(A-B)}{2}}{\cos A} \right] &= b \left[\frac{\sin \frac{(A-B)}{2}}{\cos B} \right] \\ \Rightarrow (k \sin A) \left[\frac{\sin \left(\frac{A-B}{2} \right)}{\cos A} \right] &= (k \sin B) \left[\frac{\sin \left(\frac{A-B}{2} \right)}{\cos B} \right] \\ \left\{ \begin{array}{l} \text{In } \triangle ABC \\ a = k \sin A \\ b = k \sin B \end{array} \right\} \\ \Rightarrow \tan A &= \tan B \\ \Rightarrow \angle A = \angle B &\Rightarrow A = B\end{aligned}$$

Question73

In $\triangle ABC$, medians AD and BE are drawn. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$, then the area of $\triangle ABC$ is

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Options:

A. $\frac{8}{3}$ sq units

B. $\frac{16}{3}$ sq units

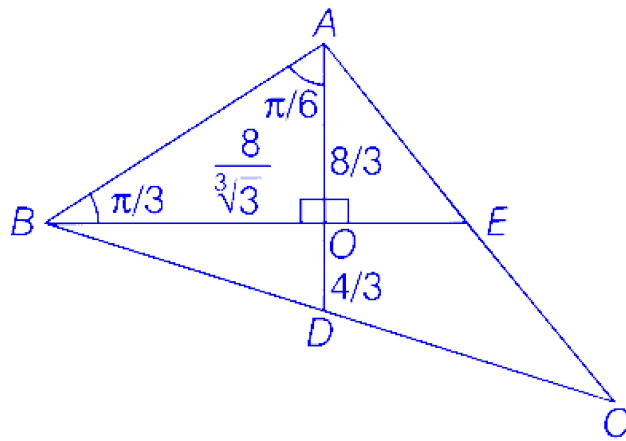
C.

$$\frac{32}{3\sqrt{3}} \text{ sq. units}$$

D. $\frac{64}{3}$ sq units

Answer: C

Solution:



Given : $AD = 4$; $\angle DAB = \frac{\pi}{6}$; $\angle ABE = \frac{\pi}{3}$

$$AO = \frac{8}{3} \text{ and } OD = \frac{4}{3}$$

$$[\because O \text{ divides } AD \text{ in } 2 : 1]$$

$$\text{Area of } \triangle ABC = 2(\text{Area of } \triangle ABE)$$

$$= 2 \left[\frac{3}{2} (\text{Area of } \triangle AOB) \right]$$

$$= 3 \times \text{area of } \triangle AOB$$

$$= 3 \times \frac{1}{2} \times BO \times AO$$

$$= 3 \cdot \frac{1}{2} \cdot \frac{8}{3\sqrt{3}} \cdot \frac{8}{3}$$

$$\left\{ \because \tan \frac{\pi}{6} = \frac{BO}{AO} \Rightarrow BO = \frac{1}{\sqrt{3}} \times \frac{8}{3} \right\}$$

$$= \frac{32}{3\sqrt{3}}$$

Question 74

In a $\triangle ABC$, $2\Delta^2 = \frac{a^2b^2c^2}{a^2+b^2+c^2}$, then the triangle is

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Options:

- A. equilateral
- B. isosceles
- C. right angled
- D. acute angled triangle

Answer: C

Solution:

$$2\Delta^2 (a^2 + b^2 + c^2) = a^2b^2c^2$$
$$a^2 + b^2 + c^2 = \left(\frac{abc}{\Delta}\right)^2 \cdot \frac{1}{2} = 8R^2$$

[$\because abc/\Delta = 4R$ where R -circumradius

$$\begin{aligned} \because a &= 2R \sin A, b = 2R \sin B, c = 2R \sin C \\ \Rightarrow 4R^2 (\sin^2 A + \sin^2 B + \sin^2 C) &= 8R^2 \\ \Rightarrow \sin^2 A + \sin^2 B + \sin^2 C &= 2 \\ \Rightarrow (\cos 2A + \cos 2B + \cos 2C) &= -1 \\ \Rightarrow -1 - 4 \cos A \cos B \cos C &= -1 \\ \cos A \cos B \cos C &= 0 \end{aligned}$$

So, any one among A , B and C has to be $\frac{\pi}{2}$.

So, right angled triangle.

Question 75

In $\triangle ABC$, suppose the radius of the circle opposite to an angle A is denoted by r_1 , similarly $r_2 \leftrightarrow \text{angle } B$, $r_3 \leftrightarrow \text{angle } C$. If $r_1 = 2, r_2 = 3, r_3 = 6$, what is the value of $r_1 + r_2 + r_3 - r = (\mathbf{R - radius of the circum circle})$.

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Options:

A. $4R$

B. $3R$

C. $2R$

D. R

Answer: A

Solution:

$$\because r = \frac{\Delta}{s}, r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b},$$

$$r_3 = \frac{\Delta}{s-c} \text{ and } 2s = a + b + c$$

$$\text{Then, } r_1 + r_2 + r_3 - r = \frac{\Delta}{s-a} + \frac{\Delta}{s-b} + \frac{\Delta}{s-c} - \frac{\Delta}{s}$$

$$= \Delta \left(\frac{2s - (a+b)}{(s-a)(s-b)} \right) + \Delta \left(\frac{c}{s(s-c)} \right)$$

$$= \Delta \frac{(2s - 2s + c)}{(s-a)(s-b)} + \Delta \left(\frac{c}{s(s-c)} \right)$$

$$= \Delta \left[\frac{cs(s-c) + c(s-a)(s-b)}{s(s-a)(s-b)(s-c)} \right]$$

$$[\because \Delta^2 = s(s-a)(s-b)(s-c)]$$

$$= \frac{\Delta c}{\Delta^2} (s(s-c) + (s-a)(s-b))$$

$$= \frac{c}{\Delta} (s^2 - sc + s^2 - (a+b)s + ab)$$

$$= \frac{c}{\Delta} (2s^2 - (a+b+c)s + ab)$$

$$= \frac{c}{\Delta} (2s^2 - 2s^2 + ab) [\because 2s = a + b + c]$$

$$= \frac{abc}{\Delta} = 4 \left(\frac{abc}{4\Delta} \right) = 4R$$

Question 76

In a $\triangle ABC$, if $a = 3$, $b = 4$ and $\sin A = \frac{3}{4}$, then $\angle CBA$ is equal to

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Options:

A. 60°

B. 75°

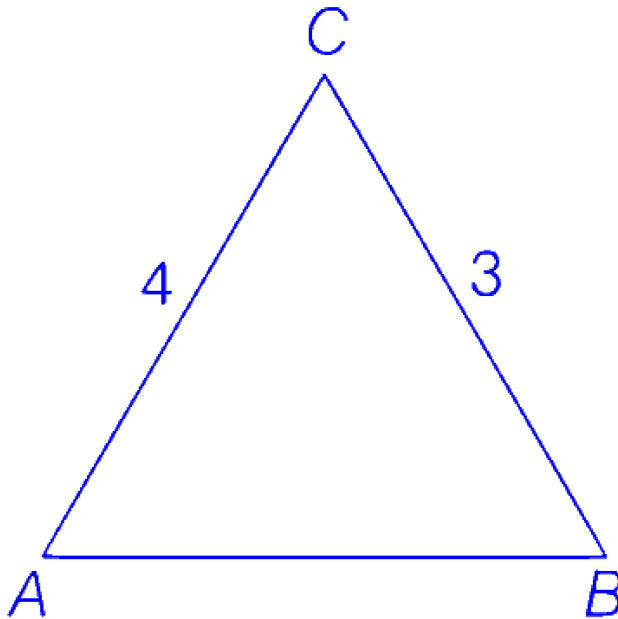
C. 90°

D. 45°

Answer: C

Solution:

In $\triangle ABC$, $a = 3$, $b = 4$, $\sin A = \frac{3}{4}$



$$\therefore \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$$

$$\text{But } \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow \frac{\sqrt{7}}{4} = \frac{4^2 + c^2 - 3^2}{2 \cdot 4 \cdot c}$$

$$\Rightarrow 2\sqrt{7}c = 7 + c^2$$

$$\Rightarrow c^2 - 2\sqrt{7}c + 7 = 0$$

$$\Rightarrow (c - \sqrt{7})^2 = 0$$

$$\therefore c = \sqrt{7}$$

$$\text{Now, } \cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

$$\Rightarrow \cos B = \frac{7+9-16}{2 \cdot 3 \cdot \sqrt{7}} = 0 \Rightarrow B = 90^\circ$$

$$\therefore \angle CBA = 90^\circ$$

Question 77

In $\triangle ABC$, $A = 75^\circ$ and $B = 45^\circ$, then the value of $b + c\sqrt{2}$ is equal to

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Options:

A. a

B. 3a

C. 2a

D. 4a

Answer: C

Solution:

$$A = 75^\circ, B = 45^\circ, \text{ then } C = 180^\circ - (75^\circ + 45^\circ) = 60^\circ$$

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \text{ where } R \text{ is circumradius}$$

$$\Rightarrow a = 2R \sin A, b = 2R \sin B, c = 2R \sin C$$

$$\text{Then, } b + c\sqrt{2} = 2R(\sin B + \sqrt{2} \sin C)$$

$$\begin{aligned}
&= 2R(\sin 45Y + \sqrt{2} \sin 60Y) \\
&= 2R \left(\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}} \right) = 2R \left(\frac{\sqrt{3} + 1}{\sqrt{2}} \right) \\
&= 2 \cdot 2R \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right) \\
&= 2 \cdot (2R \sin 75Y) \\
&= 2(2R \sin A) \\
&= 2a
\end{aligned}$$

Question 78

In $\triangle ABC$, suppose the radius of the circle opposite to an $\angle A$ is denoted by r_1 , similarly $r_2 \leftrightarrow \angle B$ and $r_3 \leftrightarrow \angle C$. If r is the radius of inscribed circle, then, what is the value of $\frac{ab - r_1 r_2}{r_3}$ is equal to

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Options:

A. $r_1 r_2 r_3$

B. r

C. $r_1 r_2 \frac{r_3}{2}$

D. $\frac{r}{2}$

Answer: B

Solution:

In $\triangle ABC$

$$a = 2R \sin A = 4R \sin \frac{A}{2} \cos \frac{A}{2}$$

$$b = 2R \sin B = 4R \sin \frac{B}{2} \cos \frac{B}{2}$$

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$r_2 = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}$$

$$r_3 = 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

$$ab - r_1 r_2 = 16R^2 \sin \frac{A}{2} \sin \frac{B}{2} \cos \frac{A}{2}$$

$$\text{Now, } \cos \frac{B}{2} \left(1 - \cos^2 \frac{C}{2} \right)$$

$$= 16R^2 \sin \frac{A}{2} \sin \frac{B}{2} \cos \frac{A}{2} \cos \frac{B}{2} \sin^2 \frac{C}{2}$$

$$\therefore \frac{ab - r_1 r_2}{r_3} = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = r$$

Question 79

If D, E and F are respectively mid-points of AB, AC and BC in $\triangle ABC$, then $BE + AF$ is equal to

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Options:

A. DC

B. $\frac{3}{2}BF$

C. $\frac{1}{2}BF$

D. $\frac{1}{2}DC$

Answer: A

Solution:

Let A, B, C is represented as **a**, **b**, **c** respectively.

Mid-points D, E, F is $\frac{\mathbf{a}+\mathbf{b}}{2}, \frac{\mathbf{a}+\mathbf{c}}{2}, \frac{\mathbf{b}+\mathbf{c}}{2}$ respectively.

$$\text{Then, } \mathbf{BE} = \frac{\mathbf{a} - 2\mathbf{b} + \mathbf{c}}{2}, \mathbf{AF} = \frac{\mathbf{b} - 2\mathbf{a} + \mathbf{c}}{2},$$

$$\mathbf{CD} = \frac{\mathbf{a} + \mathbf{b} - 2\mathbf{c}}{2}$$

$$\begin{aligned}\mathbf{BE} + \mathbf{AF} &= \frac{2\mathbf{c} - \mathbf{a} - \mathbf{b}}{2} = \frac{-(\mathbf{a} + \mathbf{b} - 2\mathbf{c})}{2} \\ &= -\mathbf{CD} = \mathbf{DC}\end{aligned}$$
